

India District-Level Soil Dataset

Soil Fertility, Moisture, pH, Nitrogen & Phosphorus by State and District

This document is formatted as one self-contained descriptive paragraph per district, so it can be directly chunked and indexed by a Retrieval-Augmented Generation (RAG) pipeline. Each paragraph repeats the state and district name along with all soil parameters, so a single retrieved chunk is fully interpretable without needing neighbouring context.

1. What This Dataset Contains

This dataset covers 640 districts across 35 Indian states and union territories (administrative boundaries as per Census of India 2011). For each district it records: dominant Soil Type, Soil Fertility (%), Soil Moisture (%), Soil pH, available Nitrogen (kg/ha), available Phosphorus (kg/ha), and a derived Fertility Class (Low/Medium/High).

2. Data Sources and Method

District and state names are taken from the public Census of India 2011 district list. Note that some states have since been reorganised (for example Telangana was carved out of Andhra Pradesh in 2014, and several new districts have been notified between 2011 and 2026); this dataset reflects 2011-era administrative boundaries.

Soil Type for each district is assigned from its state's dominant soil order as classified by ICAR and the National Bureau of Soil Survey & Land Use Planning (NBSS-LUP) - e.g. Alluvial soils across the Indo-Gangetic plains, Black (Regur) soils across the Deccan trap region, Red and Laterite soils across the peninsular plateau and Western Ghats, Arid soils across Rajasthan, and Mountain/Forest soils across the Himalayan states. This classification is real, published soil science and is not fabricated.

Soil Fertility %, Soil Moisture %, Soil pH, Nitrogen (kg/ha) and Phosphorus (kg/ha) are **indicative regional estimates**. They were generated within realistic value ranges published for each soil type, informed by NBSS-LUP soil survey reports and by publicly reported Soil Health Card (SHC) scheme cycle-wise averages (for example, published SHC case studies report soil pH in the 6.5-7.9 band, available nitrogen in roughly the 100-400 kg/ha band, and available phosphorus in roughly the 5-25 kg/ha band, depending on soil type and region). **These figures are not individual laboratory measurements for each specific district** - genuine, per-district lab data is generated per farm holding under India's Soil Health Card scheme and is not published anywhere as a single clean, bulk, district-summary file that can be reliably scraped in bulk. Presenting invented precision as if it were scraped lab data would be misleading, so every value here is explicitly labelled as an indicative estimate.

For genuine lab-grade, per-farm soil test data, consult: the Soil Health Card portal (soilhealth.dac.gov.in); ICAR-NBSS&LUP district soil survey reports; or your State Agriculture Department / local Krishi Vigyan Kendra.

Weather / climate columns (Climate Zone, Average Annual Rainfall, Average Temperature Range, Seasonal Weather Pattern) are assigned per state from published Koppen-Geiger climate classification and India Meteorological Department (IMD) climatic sub-division norms. These are **long-term climatology (typical seasonal norms)**, not live or real-time weather. For current forecasts and warnings, use IMD directly (mausam.imd.gov.in).

Crop Suitability is derived from standard ICAR/agronomy guidance on which crops are typically grown on each soil type (e.g. Alluvial soil suits wheat, rice, sugarcane and maize; Black/Regur soil suits cotton and soybean; Laterite soil suits rice, coconut, cashew, tea and spices), adjusted for the district's rainfall band. This is a general regional guide, not a field-specific agronomic recommendation.

3. How to Use This With a RAG Pipeline

Section 5 below contains one paragraph per district, grouped by state under clear headings. Each paragraph is self-contained (repeats state, district, and all metrics), making it suitable as a single retrieval chunk (roughly 60-90 tokens each) without further splitting. The accompanying India_Soil_Dataset.xlsx file contains the same data in structured tabular form for filtering, joins, or use as a lookup table alongside this PDF.

4. Soil Type Reference Table

Soil Type	Typical Region	Fertility %	Moisture %	pH	N (kg/ha)	P (kg/ha)
Alluvial Soil	Indo-Gangetic plains	55-80	18-32	6.5-7.8	180-320	12-28
Black Soil (Regur)	Deccan trap region	50-75	20-35	7.0-8.3	150-280	8-20
Red & Black Soil	AP, Karnataka, TN	45-68	14-26	6.2-7.5	140-250	9-22
Red & Yellow Soil	Jharkhand, Chhattisgarh	38-60	12-22	5.5-6.8	110-220	6-16
Red & Laterite Soil	Odisha	35-58	15-28	5.0-6.5	100-210	5-14
Red & Lateritic Soil	NE hill states	35-58	18-32	4.8-6.2	100-200	5-14
Laterite Soil	Kerala, Goa	32-55	20-35	4.5-5.8	90-190	4-12
Arid & Desert Soil	Rajasthan	18-38	3-10	7.5-8.8	40-110	3-10
Mountain & Forest Soil	Himalayan states	40-65	22-38	5.0-6.8	120-240	6-16
Coastal Alluvial Soil	Puducherry, A&N Islands	42-66	24-40	6.8-8.0	130-240	7-18
Coastal Sandy Soil	Lakshadweep	25-42	15-26	6.5-7.8	60-140	4-10

4b. Climate Zone Reference (by State)

State	Climate Zone	Rainfall (mm/yr)	Temp Range (C)
Andaman & Nicobar Islands	Tropical Monsoon (Koppen Am)	3000-3500	23 to 31
Andhra Pradesh	Tropical Savanna (Koppen Aw)	900-1100	14 to 42
Arunachal Pradesh	Humid Subtropical / Highland (Koppen Cwa/Cwb)	2000-4500	2 to 32
Assam	Humid Subtropical (Koppen Cwa)	1800-3200	7 to 35
Bihar	Humid Subtropical (Koppen Cwa)	1000-1600	7 to 42
Chandigarh	Semi-Arid Subtropical (Koppen BSh)	700-1100	5 to 41
Chhattisgarh	Tropical Wet & Dry (Koppen Aw)	1200-1600	8 to 44
Dadra & Nagar Haveli	Tropical Wet & Dry (Koppen Aw)	1900-2400	14 to 36
Daman & Diu	Tropical Wet & Dry (Koppen Aw)	1800-2200	16 to 34
Delhi	Semi-Arid Subtropical (Koppen BSh)	600-800	3 to 45
Goa	Tropical Monsoon (Koppen Am)	2500-3200	19 to 35
Gujarat	Semi-Arid to Arid (Koppen BSh/BWh)	300-2000	7 to 43
Haryana	Semi-Arid Subtropical (Koppen BSh)	350-800	4 to 44
Himachal Pradesh	Humid Subtropical / Highland (Koppen Cwb)	1000-1800	(-2) to 32
Jammu And Kashmir	Cold / Alpine (Koppen Dfb/ET)	650-1200	(-5) to 30
Jharkhand	Humid Subtropical (Koppen Cwa)	1100-1500	6 to 42
Karnataka	Tropical Wet & Dry (Koppen Aw/Am)	600-3800	11 to 38
Kerala	Tropical Monsoon (Koppen Am)	2500-3500	21 to 35
Lakshadweep	Tropical Monsoon (Koppen Am)	1600-1800	22 to 32
Madhya Pradesh	Tropical Wet & Dry (Koppen Aw)	800-1400	4 to 45
Maharashtra	Tropical Wet & Dry (Koppen Aw)	600-3500	9 to 42
Manipur	Humid Subtropical (Koppen Cwa)	1400-2500	2 to 32
Meghalaya	Humid Subtropical (Koppen Cwb)	2500-11000	4 to 28
Mizoram	Humid Subtropical (Koppen Cwa)	2000-3000	7 to 30
Nagaland	Humid Subtropical (Koppen Cwa)	1800-2500	4 to 31
Odisha	Tropical Savanna (Koppen Aw)	1200-1600	12 to 40
Pondicherry	Tropical Savanna (Koppen Aw)	1200-1400	20 to 38
Punjab	Semi-Arid Subtropical (Koppen BSh)	400-900	4 to 42
Rajasthan	Arid / Desert (Koppen BWh)	100-650	2 to 48
Sikkim	Highland / Subtropical Highland (Koppen Cwb)	2000-3500	(-5) to 28
Tamil Nadu	Tropical Savanna (Koppen Aw)	700-1200	18 to 40
Tripura	Tropical Savanna (Koppen Aw)	2000-2500	10 to 35
Uttar Pradesh	Humid Subtropical (Koppen Cwa)	700-1400	4 to 45
Uttarakhand	Humid Subtropical / Highland (Koppen Cwa/ET)	1000-2000	0 to 35
West Bengal	Tropical Wet & Dry (Koppen Aw/Cwa)	1200-3000	9 to 40

5. District-Wise Soil Profiles

Each entry below is a standalone paragraph: State, District, Soil Type, Fertility %, Moisture %, pH, Nitrogen, Phosphorus, Fertility Class, Climate Zone, Rainfall, Temperature Range, Seasonal Weather Pattern and Crop Suitability. Values are indicative regional estimates as described in Section 2.

Andaman & Nicobar Islands

Nicobars district, Andaman & Nicobar Islands. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 60.6% (High fertility class). Soil moisture content is estimated at 31.0%. Soil pH is estimated at 7.83 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 8.0 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 3000-3500 mm and an average temperature range of 23 to 31 degrees Celsius. Seasonal pattern: Warm humid maritime climate year-round, two monsoons. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North And Middle Andaman district, Andaman & Nicobar Islands. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 65.4% (High fertility class). Soil moisture content is estimated at 36.2%. Soil pH is estimated at 7.74 (on a scale of 0-14). Available nitrogen is estimated at 144 kg/ha, and available phosphorus is estimated at 12.0 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 3000-3500 mm and an average temperature range of 23 to 31 degrees Celsius. Seasonal pattern: Warm humid maritime climate year-round, two monsoons. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South Andaman district, Andaman & Nicobar Islands. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 50.9% (Medium fertility class). Soil moisture content is estimated at 38.8%. Soil pH is estimated at 7.57 (on a scale of 0-14). Available nitrogen is estimated at 220 kg/ha, and available phosphorus is estimated at 11.9 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 3000-3500 mm and an average temperature range of 23 to 31 degrees Celsius. Seasonal pattern: Warm humid maritime climate year-round, two monsoons. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Andhra Pradesh

Adilabad district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 50.2% (Medium fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 6.28 (on a scale of 0-14). Available nitrogen is estimated at 231 kg/ha, and available phosphorus is estimated at 17.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Anantapur district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 62.4% (High fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 7.46 (on a scale of 0-14). Available nitrogen is estimated at 238 kg/ha, and available phosphorus is estimated at 19.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chittoor district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.5% (Medium fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 6.26 (on a scale of 0-14). Available nitrogen is estimated at 156 kg/ha, and available phosphorus is estimated at 17.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

East Godavari district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 62.1% (High fertility class). Soil moisture content is estimated at 25.6%. Soil pH is estimated at 6.62 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 15.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Guntur district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.4% (Medium fertility class). Soil moisture content is estimated at 15.6%. Soil pH is estimated at 6.82 (on a scale of 0-14). Available nitrogen is estimated at 164 kg/ha, and available phosphorus is estimated at 17.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hyderabad district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 55.1% (High fertility class). Soil moisture content is estimated at 24.0%. Soil pH is estimated at 7.11 (on a scale of 0-14). Available nitrogen is estimated at 174 kg/ha, and available phosphorus is estimated at 19.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karimnagar district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 63.5% (High fertility class). Soil moisture content is estimated at 18.6%. Soil pH is estimated at 6.57 (on a

scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 10.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khammam district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.6% (Medium fertility class). Soil moisture content is estimated at 14.1%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 18.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Krishna district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 63.0% (High fertility class). Soil moisture content is estimated at 19.5%. Soil pH is estimated at 6.94 (on a scale of 0-14). Available nitrogen is estimated at 155 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kurnool district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 60.4% (High fertility class). Soil moisture content is estimated at 19.7%. Soil pH is estimated at 6.93 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 17.3 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahbubnagar district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 57.7% (High fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 6.6 (on a scale of 0-14). Available nitrogen is estimated at 143 kg/ha, and available phosphorus is estimated at 14.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Medak district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.9% (Medium fertility class). Soil moisture content is estimated at 18.9%. Soil pH is estimated at 7.31 (on a scale of 0-14). Available nitrogen is estimated at 165 kg/ha, and available phosphorus is estimated at 9.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of

900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nalgonda district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 51.5% (Medium fertility class). Soil moisture content is estimated at 17.5%. Soil pH is estimated at 7.06 (on a scale of 0-14). Available nitrogen is estimated at 201 kg/ha, and available phosphorus is estimated at 19.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nizamabad district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 60.3% (High fertility class). Soil moisture content is estimated at 18.9%. Soil pH is estimated at 7.26 (on a scale of 0-14). Available nitrogen is estimated at 158 kg/ha, and available phosphorus is estimated at 9.3 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Prakasam district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 47.1% (Medium fertility class). Soil moisture content is estimated at 22.7%. Soil pH is estimated at 6.8 (on a scale of 0-14). Available nitrogen is estimated at 157 kg/ha, and available phosphorus is estimated at 15.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rangareddy district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 48.5% (Medium fertility class). Soil moisture content is estimated at 22.4%. Soil pH is estimated at 6.78 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 12.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sri Potti Sriramulu Nellore district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 59.5% (High fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 6.31 (on a scale of 0-14). Available nitrogen is estimated at 152 kg/ha, and available phosphorus is estimated at 21.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE

monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Srikakulam district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.9% (High fertility class). Soil moisture content is estimated at 22.4%. Soil pH is estimated at 6.55 (on a scale of 0-14). Available nitrogen is estimated at 246 kg/ha, and available phosphorus is estimated at 19.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Visakhapatnam district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 61.5% (High fertility class). Soil moisture content is estimated at 19.4%. Soil pH is estimated at 6.55 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 20.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Vizianagaram district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 55.5% (High fertility class). Soil moisture content is estimated at 16.4%. Soil pH is estimated at 6.6 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Warangal district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.7% (High fertility class). Soil moisture content is estimated at 23.1%. Soil pH is estimated at 7.14 (on a scale of 0-14). Available nitrogen is estimated at 187 kg/ha, and available phosphorus is estimated at 17.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Godavari district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 58.4% (High fertility class). Soil moisture content is estimated at 21.8%. Soil pH is estimated at 6.31 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

measurement or live weather reading.]

Y.S.R. district, Andhra Pradesh. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 18.0%. Soil pH is estimated at 6.39 (on a scale of 0-14). Available nitrogen is estimated at 151 kg/ha, and available phosphorus is estimated at 16.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 900-1100 mm and an average temperature range of 14 to 42 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep and Oct-Dec (NE monsoon). Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Arunachal Pradesh

Anjaw district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 44.3% (Medium fertility class). Soil moisture content is estimated at 36.8%. Soil pH is estimated at 6.05 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 11.9 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Changlang district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 40.6% (Medium fertility class). Soil moisture content is estimated at 37.3%. Soil pH is estimated at 5.87 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 6.8 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dibang Valley district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 52.2% (Medium fertility class). Soil moisture content is estimated at 29.9%. Soil pH is estimated at 6.69 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 10.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

East Kameng district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.7% (Medium fertility class). Soil moisture content is estimated at 27.3%. Soil pH is estimated at 5.94 (on a scale of 0-14). Available nitrogen is estimated at 172 kg/ha, and available phosphorus is estimated at 6.2 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen

Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

East Siang district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 60.7% (High fertility class). Soil moisture content is estimated at 36.3%. Soil pH is estimated at 5.25 (on a scale of 0-14). Available nitrogen is estimated at 186 kg/ha, and available phosphorus is estimated at 7.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kurung Kumey district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.8% (High fertility class). Soil moisture content is estimated at 26.5%. Soil pH is estimated at 6.19 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lohit district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 42.7% (Medium fertility class). Soil moisture content is estimated at 36.7%. Soil pH is estimated at 5.41 (on a scale of 0-14). Available nitrogen is estimated at 124 kg/ha, and available phosphorus is estimated at 11.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lower Dibang Valley district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 49.3% (Medium fertility class). Soil moisture content is estimated at 35.3%. Soil pH is estimated at 6.45 (on a scale of 0-14). Available nitrogen is estimated at 158 kg/ha, and available phosphorus is estimated at 15.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lower Subansiri district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 47.3% (Medium fertility class). Soil moisture content is estimated at 30.2%. Soil pH is estimated at 5.46 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 7.6 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Papum Pare district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 41.1% (Medium fertility class). Soil moisture content is estimated at 29.0%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 227 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tawang district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 62.3% (High fertility class). Soil moisture content is estimated at 36.3%. Soil pH is estimated at 5.93 (on a scale of 0-14). Available nitrogen is estimated at 157 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tirap district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.5% (High fertility class). Soil moisture content is estimated at 28.0%. Soil pH is estimated at 5.17 (on a scale of 0-14). Available nitrogen is estimated at 209 kg/ha, and available phosphorus is estimated at 8.6 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Upper Siang district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 63.4% (High fertility class). Soil moisture content is estimated at 25.9%. Soil pH is estimated at 5.22 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 7.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative

regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Upper Subansiri district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 44.5% (Medium fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 143 kg/ha, and available phosphorus is estimated at 9.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Kameng district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 59.4% (High fertility class). Soil moisture content is estimated at 37.5%. Soil pH is estimated at 5.9 (on a scale of 0-14). Available nitrogen is estimated at 137 kg/ha, and available phosphorus is estimated at 6.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Siang district, Arunachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 45.7% (Medium fertility class). Soil moisture content is estimated at 24.1%. Soil pH is estimated at 6.22 (on a scale of 0-14). Available nitrogen is estimated at 134 kg/ha, and available phosphorus is estimated at 11.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/Cwb) climate zone, with typical annual rainfall of 2000-4500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Heavy monsoon rainfall, cool winters in hills. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Assam

Baksa district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.4% (High fertility class). Soil moisture content is estimated at 26.1%. Soil pH is estimated at 6.76 (on a scale of 0-14). Available nitrogen is estimated at 292 kg/ha, and available phosphorus is estimated at 23.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Barpeta district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 19.8%. Soil pH is estimated at 6.66 (on a scale of 0-14). Available nitrogen is estimated at 309 kg/ha, and available phosphorus is estimated at 18.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200

mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bongaigaon district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.5% (High fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 7.36 (on a scale of 0-14). Available nitrogen is estimated at 313 kg/ha, and available phosphorus is estimated at 16.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Cachar district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.1% (High fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 268 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chirang district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.5% (High fertility class). Soil moisture content is estimated at 23.9%. Soil pH is estimated at 7.76 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 26.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Darrang district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.1% (High fertility class). Soil moisture content is estimated at 24.5%. Soil pH is estimated at 7.52 (on a scale of 0-14). Available nitrogen is estimated at 182 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhemaji district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.7% (High fertility class). Soil moisture content is estimated at 29.2%. Soil pH is estimated at 6.8 (on a scale of 0-14). Available nitrogen is estimated at 254 kg/ha, and available phosphorus is estimated at 21.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhubri district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 76.7% (High fertility class). Soil moisture content is estimated at 26.4%. Soil pH is estimated at 7.04 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 18.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dibrugarh district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 30.1%. Soil pH is estimated at 7.09 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dima Hasao district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.7% (High fertility class). Soil moisture content is estimated at 29.4%. Soil pH is estimated at 6.64 (on a scale of 0-14). Available nitrogen is estimated at 189 kg/ha, and available phosphorus is estimated at 21.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Goalpara district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.7% (High fertility class). Soil moisture content is estimated at 29.5%. Soil pH is estimated at 6.9 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 26.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Golaghat district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.1% (High fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 6.7 (on a scale of 0-14). Available nitrogen is estimated at 196 kg/ha, and available phosphorus is estimated at 12.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hailakandi district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 262 kg/ha, and available phosphorus is estimated at 22.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot

humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jorhat district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.8% (High fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 7.16 (on a scale of 0-14). Available nitrogen is estimated at 302 kg/ha, and available phosphorus is estimated at 25.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kamrup district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 20.5%. Soil pH is estimated at 6.81 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 21.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kamrup Metropolitan district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.4% (High fertility class). Soil moisture content is estimated at 18.7%. Soil pH is estimated at 6.99 (on a scale of 0-14). Available nitrogen is estimated at 253 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karbi Anglong district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.8% (High fertility class). Soil moisture content is estimated at 18.7%. Soil pH is estimated at 7.7 (on a scale of 0-14). Available nitrogen is estimated at 193 kg/ha, and available phosphorus is estimated at 25.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karimganj district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.6% (High fertility class). Soil moisture content is estimated at 31.7%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 289 kg/ha, and available phosphorus is estimated at 22.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kokrajhar district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.5% (High fertility class). Soil moisture content is estimated at 19.9%. Soil pH is estimated at 7.2 (on a scale of 0-14). Available nitrogen is estimated at 251 kg/ha, and available phosphorus is estimated at 25.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lakhimpur district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.6% (High fertility class). Soil moisture content is estimated at 23.4%. Soil pH is estimated at 7.33 (on a scale of 0-14). Available nitrogen is estimated at 217 kg/ha, and available phosphorus is estimated at 14.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Morigaon district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.9% (High fertility class). Soil moisture content is estimated at 23.8%. Soil pH is estimated at 6.8 (on a scale of 0-14). Available nitrogen is estimated at 231 kg/ha, and available phosphorus is estimated at 17.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nagaon district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.2% (High fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 7.39 (on a scale of 0-14). Available nitrogen is estimated at 221 kg/ha, and available phosphorus is estimated at 27.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nalbari district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.9% (High fertility class). Soil moisture content is estimated at 24.7%. Soil pH is estimated at 6.93 (on a scale of 0-14). Available nitrogen is estimated at 254 kg/ha, and available phosphorus is estimated at 25.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sivasagar district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 7.19 (on a scale of 0-14). Available nitrogen is estimated at 268 kg/ha, and available phosphorus is estimated at 16.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination

include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sonitpur district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.4% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 7.4 (on a scale of 0-14). Available nitrogen is estimated at 300 kg/ha, and available phosphorus is estimated at 14.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tinsukia district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.4% (High fertility class). Soil moisture content is estimated at 19.3%. Soil pH is estimated at 7.44 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 27.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Udalguri district, Assam. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.4% (High fertility class). Soil moisture content is estimated at 31.4%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 24.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-3200 mm and an average temperature range of 7 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, heavy monsoon. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bihar

Araria district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.9% (High fertility class). Soil moisture content is estimated at 31.2%. Soil pH is estimated at 6.83 (on a scale of 0-14). Available nitrogen is estimated at 262 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Arwal district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.4% (High fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 7.23 (on a scale of 0-14). Available nitrogen is estimated at 260 kg/ha, and available phosphorus is estimated at 19.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Aurangabad district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.1% (High fertility class). Soil moisture content is estimated at 28.7%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 248 kg/ha, and available phosphorus is estimated at 21.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Banka district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.3% (High fertility class). Soil moisture content is estimated at 19.7%. Soil pH is estimated at 6.65 (on a scale of 0-14). Available nitrogen is estimated at 192 kg/ha, and available phosphorus is estimated at 22.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Begusarai district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.5% (High fertility class). Soil moisture content is estimated at 28.8%. Soil pH is estimated at 7.37 (on a scale of 0-14). Available nitrogen is estimated at 226 kg/ha, and available phosphorus is estimated at 26.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhagalpur district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.1% (High fertility class). Soil moisture content is estimated at 21.8%. Soil pH is estimated at 6.97 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 14.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhojpur district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.7% (High fertility class). Soil moisture content is estimated at 31.1%. Soil pH is estimated at 7.07 (on a scale of 0-14). Available nitrogen is estimated at 233 kg/ha, and available phosphorus is estimated at 23.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Buxar district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.8% (High fertility class). Soil moisture content is estimated at 31.1%. Soil pH is estimated at 7.51 (on a scale of 0-14). Available nitrogen is estimated at 247 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat,

rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Darbhanga district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.7% (High fertility class). Soil moisture content is estimated at 28.0%. Soil pH is estimated at 7.74 (on a scale of 0-14). Available nitrogen is estimated at 196 kg/ha, and available phosphorus is estimated at 25.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gaya district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.9% (High fertility class). Soil moisture content is estimated at 19.7%. Soil pH is estimated at 7.26 (on a scale of 0-14). Available nitrogen is estimated at 276 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gopalganj district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.4% (High fertility class). Soil moisture content is estimated at 29.6%. Soil pH is estimated at 6.88 (on a scale of 0-14). Available nitrogen is estimated at 244 kg/ha, and available phosphorus is estimated at 19.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jamui district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.5% (High fertility class). Soil moisture content is estimated at 30.9%. Soil pH is estimated at 7.52 (on a scale of 0-14). Available nitrogen is estimated at 195 kg/ha, and available phosphorus is estimated at 28.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jehanabad district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.0% (High fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 7.59 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 28.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kaimur (Bhabua) district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.6% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 6.62 (on a scale of

0-14). Available nitrogen is estimated at 305 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Katihar district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.0% (High fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 7.51 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 26.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khagaria district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.4% (High fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 6.51 (on a scale of 0-14). Available nitrogen is estimated at 187 kg/ha, and available phosphorus is estimated at 21.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kishanganj district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.0% (High fertility class). Soil moisture content is estimated at 21.3%. Soil pH is estimated at 7.6 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 24.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lakhisarai district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.2% (High fertility class). Soil moisture content is estimated at 28.8%. Soil pH is estimated at 7.41 (on a scale of 0-14). Available nitrogen is estimated at 297 kg/ha, and available phosphorus is estimated at 12.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Madhepura district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.0% (High fertility class). Soil moisture content is estimated at 19.7%. Soil pH is estimated at 7.16 (on a scale of 0-14). Available nitrogen is estimated at 284 kg/ha, and available phosphorus is estimated at 22.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

measurement or live weather reading.]

Madhubani district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 76.3% (High fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 7.46 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 16.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Munger district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.7% (High fertility class). Soil moisture content is estimated at 31.7%. Soil pH is estimated at 7.3 (on a scale of 0-14). Available nitrogen is estimated at 187 kg/ha, and available phosphorus is estimated at 21.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Muzaffarpur district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 30.4%. Soil pH is estimated at 7.42 (on a scale of 0-14). Available nitrogen is estimated at 204 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nalanda district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.6% (High fertility class). Soil moisture content is estimated at 31.7%. Soil pH is estimated at 7.1 (on a scale of 0-14). Available nitrogen is estimated at 289 kg/ha, and available phosphorus is estimated at 22.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nawada district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.3% (High fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 7.73 (on a scale of 0-14). Available nitrogen is estimated at 222 kg/ha, and available phosphorus is estimated at 21.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pashchim Champaran district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.5% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 20.1 kg/ha.

The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Patna district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.1% (High fertility class). Soil moisture content is estimated at 24.6%. Soil pH is estimated at 6.65 (on a scale of 0-14). Available nitrogen is estimated at 198 kg/ha, and available phosphorus is estimated at 16.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Purba Champaran district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.6% (High fertility class). Soil moisture content is estimated at 24.0%. Soil pH is estimated at 7.29 (on a scale of 0-14). Available nitrogen is estimated at 268 kg/ha, and available phosphorus is estimated at 18.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Purnia district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.2% (High fertility class). Soil moisture content is estimated at 21.0%. Soil pH is estimated at 7.26 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 12.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rohtas district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.5% (High fertility class). Soil moisture content is estimated at 24.6%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 260 kg/ha, and available phosphorus is estimated at 21.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Saharsa district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.6% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 7.35 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 24.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Samastipur district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.7% (High fertility class). Soil moisture content is estimated at 24.0%. Soil pH is estimated at 7.31 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 20.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Saran district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.4% (High fertility class). Soil moisture content is estimated at 30.4%. Soil pH is estimated at 7.7 (on a scale of 0-14). Available nitrogen is estimated at 250 kg/ha, and available phosphorus is estimated at 20.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sheikhpura district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.0% (High fertility class). Soil moisture content is estimated at 22.4%. Soil pH is estimated at 7.59 (on a scale of 0-14). Available nitrogen is estimated at 249 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sheohar district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.8% (High fertility class). Soil moisture content is estimated at 29.8%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 17.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sitamarhi district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.3% (High fertility class). Soil moisture content is estimated at 22.4%. Soil pH is estimated at 7.47 (on a scale of 0-14). Available nitrogen is estimated at 182 kg/ha, and available phosphorus is estimated at 25.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Siwan district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 76.4% (High fertility class). Soil moisture content is estimated at 23.2%. Soil pH is estimated at 6.7 (on a scale of 0-14). Available nitrogen is estimated at 264 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers,

mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Supaul district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.1% (High fertility class). Soil moisture content is estimated at 31.4%. Soil pH is estimated at 7.79 (on a scale of 0-14). Available nitrogen is estimated at 288 kg/ha, and available phosphorus is estimated at 17.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Vaishali district, Bihar. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.2% (High fertility class). Soil moisture content is estimated at 27.9%. Soil pH is estimated at 7.0 (on a scale of 0-14). Available nitrogen is estimated at 269 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1000-1600 mm and an average temperature range of 7 to 42 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chandigarh

Chandigarh district, Chandigarh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.2% (High fertility class). Soil moisture content is estimated at 25.4%. Soil pH is estimated at 6.71 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 25.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 700-1100 mm and an average temperature range of 5 to 41 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chhattisgarh

Bastar district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 59.8% (High fertility class). Soil moisture content is estimated at 17.6%. Soil pH is estimated at 6.59 (on a scale of 0-14). Available nitrogen is estimated at 218 kg/ha, and available phosphorus is estimated at 7.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bijapur district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 47.9% (Medium fertility class). Soil moisture content is estimated at 15.9%. Soil pH is estimated at 5.6 (on a scale of 0-14). Available nitrogen is estimated at 193 kg/ha, and available phosphorus is estimated at 10.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot

summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bilaspur district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 48.3% (Medium fertility class). Soil moisture content is estimated at 13.5%. Soil pH is estimated at 5.74 (on a scale of 0-14). Available nitrogen is estimated at 209 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dakshin Bastar Dantewada district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 43.1% (Medium fertility class). Soil moisture content is estimated at 14.9%. Soil pH is estimated at 6.14 (on a scale of 0-14). Available nitrogen is estimated at 174 kg/ha, and available phosphorus is estimated at 10.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhamtari district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 39.9% (Medium fertility class). Soil moisture content is estimated at 14.4%. Soil pH is estimated at 6.6 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Durg district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 52.7% (Medium fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 5.58 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 11.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Janjgir - Champa district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 45.4% (Medium fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 6.13 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 14.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jashpur district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 49.1% (Medium fertility class). Soil moisture content is estimated at 21.1%. Soil pH is estimated at 6.26 (on a

scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kabeerdham district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 49.0% (Medium fertility class). Soil moisture content is estimated at 17.3%. Soil pH is estimated at 5.64 (on a scale of 0-14). Available nitrogen is estimated at 153 kg/ha, and available phosphorus is estimated at 15.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Korba district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 51.9% (Medium fertility class). Soil moisture content is estimated at 13.8%. Soil pH is estimated at 5.94 (on a scale of 0-14). Available nitrogen is estimated at 131 kg/ha, and available phosphorus is estimated at 6.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Koriya district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 58.4% (High fertility class). Soil moisture content is estimated at 16.5%. Soil pH is estimated at 5.9 (on a scale of 0-14). Available nitrogen is estimated at 175 kg/ha, and available phosphorus is estimated at 6.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahasamund district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 44.1% (Medium fertility class). Soil moisture content is estimated at 19.0%. Soil pH is estimated at 6.32 (on a scale of 0-14). Available nitrogen is estimated at 217 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Narayanpur district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 48.5% (Medium fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 6.67 (on a scale of 0-14). Available nitrogen is estimated at 189 kg/ha, and available phosphorus is estimated at 15.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Raigarh district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 55.2% (High fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 5.65 (on a scale of 0-14). Available nitrogen is estimated at 190 kg/ha, and available phosphorus is estimated at 10.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Raipur district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 50.2% (Medium fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 6.76 (on a scale of 0-14). Available nitrogen is estimated at 218 kg/ha, and available phosphorus is estimated at 8.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajnandgaon district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 54.1% (Medium fertility class). Soil moisture content is estimated at 19.5%. Soil pH is estimated at 5.95 (on a scale of 0-14). Available nitrogen is estimated at 123 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Surguja district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 55.1% (High fertility class). Soil moisture content is estimated at 16.9%. Soil pH is estimated at 6.78 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttar Bastar Kanker district, Chhattisgarh. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 47.1% (Medium fertility class). Soil moisture content is estimated at 19.9%. Soil pH is estimated at 5.61 (on a scale of 0-14). Available nitrogen is estimated at 171 kg/ha, and available phosphorus is estimated at 14.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 8 to 44 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dadra & Nagar Haveli

Dadra And Nagar Haveli district, Dadra & Nagar Haveli. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 56.3% (High fertility class). Soil moisture content is estimated at 29.5%. Soil pH is estimated at 4.85 (on a scale of 0-14). Available nitrogen is estimated at 137 kg/ha, and available

phosphorus is estimated at 5.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1900-2400 mm and an average temperature range of 14 to 36 degrees Celsius. Seasonal pattern: Hot humid, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Daman & Diu

Daman district, Daman & Diu. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 44.6% (Medium fertility class). Soil moisture content is estimated at 34.8%. Soil pH is estimated at 7.66 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1800-2200 mm and an average temperature range of 16 to 34 degrees Celsius. Seasonal pattern: Hot humid coastal climate, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Diu district, Daman & Diu. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 59.7% (High fertility class). Soil moisture content is estimated at 36.8%. Soil pH is estimated at 6.86 (on a scale of 0-14). Available nitrogen is estimated at 155 kg/ha, and available phosphorus is estimated at 13.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 1800-2200 mm and an average temperature range of 16 to 34 degrees Celsius. Seasonal pattern: Hot humid coastal climate, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Delhi

Central district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 76.5% (High fertility class). Soil moisture content is estimated at 18.1%. Soil pH is estimated at 7.17 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

East district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.0% (High fertility class). Soil moisture content is estimated at 30.5%. Soil pH is estimated at 7.37 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 25.6 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

New Delhi district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.7% (High fertility class). Soil moisture content is estimated at 29.9%. Soil pH is estimated at 6.89 (on a scale of 0-14). Available nitrogen is estimated at 262 kg/ha, and available phosphorus is estimated at 18.4 kg/ha. The

district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.9% (High fertility class). Soil moisture content is estimated at 30.4%. Soil pH is estimated at 6.74 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 17.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North East district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.9% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 7.8 (on a scale of 0-14). Available nitrogen is estimated at 221 kg/ha, and available phosphorus is estimated at 18.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North West district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.4% (High fertility class). Soil moisture content is estimated at 26.0%. Soil pH is estimated at 7.8 (on a scale of 0-14). Available nitrogen is estimated at 278 kg/ha, and available phosphorus is estimated at 21.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.8% (High fertility class). Soil moisture content is estimated at 30.8%. Soil pH is estimated at 7.15 (on a scale of 0-14). Available nitrogen is estimated at 198 kg/ha, and available phosphorus is estimated at 17.8 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South West district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.7% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 6.52 (on a scale of 0-14). Available nitrogen is estimated at 243 kg/ha, and available phosphorus is estimated at 22.2 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West district, Delhi. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.6% (High fertility class). Soil moisture content is estimated at 23.9%. Soil pH is estimated at 7.75 (on a scale of 0-14). Available nitrogen is estimated at 285 kg/ha, and available phosphorus is estimated at 20.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 600-800 mm and an average temperature range of 3 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Goa

North Goa district, Goa. Dominant soil type: Laterite Soil. Soil fertility is estimated at 38.5% (Medium fertility class). Soil moisture content is estimated at 33.5%. Soil pH is estimated at 4.81 (on a scale of 0-14). Available nitrogen is estimated at 122 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3200 mm and an average temperature range of 19 to 35 degrees Celsius. Seasonal pattern: Hot humid coastal, heavy monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South Goa district, Goa. Dominant soil type: Laterite Soil. Soil fertility is estimated at 44.2% (Medium fertility class). Soil moisture content is estimated at 31.1%. Soil pH is estimated at 5.27 (on a scale of 0-14). Available nitrogen is estimated at 155 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3200 mm and an average temperature range of 19 to 35 degrees Celsius. Seasonal pattern: Hot humid coastal, heavy monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gujarat

Ahmadabad district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.0% (High fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 7.2 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 11.4 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Amreli district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 64.0% (High fertility class). Soil moisture content is estimated at 31.9%. Soil pH is estimated at 8.02 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate

combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Anand district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 74.9% (High fertility class). Soil moisture content is estimated at 30.1%. Soil pH is estimated at 8.06 (on a scale of 0-14). Available nitrogen is estimated at 267 kg/ha, and available phosphorus is estimated at 17.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Banas Kantha district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 54.6% (Medium fertility class). Soil moisture content is estimated at 28.4%. Soil pH is estimated at 7.13 (on a scale of 0-14). Available nitrogen is estimated at 234 kg/ha, and available phosphorus is estimated at 19.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bharuch district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.8% (High fertility class). Soil moisture content is estimated at 26.5%. Soil pH is estimated at 7.05 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 9.2 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhavnagar district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.0% (Medium fertility class). Soil moisture content is estimated at 23.7%. Soil pH is estimated at 7.09 (on a scale of 0-14). Available nitrogen is estimated at 209 kg/ha, and available phosphorus is estimated at 14.2 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dohad district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.8% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 7.15 (on a scale of 0-14). Available nitrogen is estimated at 199 kg/ha, and available phosphorus is estimated at 8.7 kg/ha.

The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gandhinagar district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 67.5% (High fertility class). Soil moisture content is estimated at 23.1%. Soil pH is estimated at 7.39 (on a scale of 0-14). Available nitrogen is estimated at 201 kg/ha, and available phosphorus is estimated at 13.0 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jamnagar district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.0% (Medium fertility class). Soil moisture content is estimated at 21.7%. Soil pH is estimated at 8.12 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 14.1 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Junagadh district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.2% (High fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 7.56 (on a scale of 0-14). Available nitrogen is estimated at 251 kg/ha, and available phosphorus is estimated at 18.1 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kachchh district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.5% (High fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 7.32 (on a scale of 0-14). Available nitrogen is estimated at 212 kg/ha, and available phosphorus is estimated at 16.2 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kheda district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.7% (High fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 8.21 (on a scale of 0-14). Available nitrogen is estimated at 156 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahesana district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 67.8% (High fertility class). Soil moisture content is estimated at 22.3%. Soil pH is estimated at 7.06 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 19.0 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Narmada district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.2% (Medium fertility class). Soil moisture content is estimated at 22.8%. Soil pH is estimated at 7.04 (on a scale of 0-14). Available nitrogen is estimated at 164 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Navsari district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.0% (High fertility class). Soil moisture content is estimated at 28.5%. Soil pH is estimated at 7.77 (on a scale of 0-14). Available nitrogen is estimated at 260 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Panch Mahals district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 7.21 (on a scale of 0-14). Available nitrogen is estimated at 250 kg/ha, and available phosphorus is estimated at 18.3 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for

Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Patan district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 33.8%. Soil pH is estimated at 7.59 (on a scale of 0-14). Available nitrogen is estimated at 228 kg/ha, and available phosphorus is estimated at 19.8 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Porbandar district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 59.0% (High fertility class). Soil moisture content is estimated at 32.2%. Soil pH is estimated at 7.41 (on a scale of 0-14). Available nitrogen is estimated at 253 kg/ha, and available phosphorus is estimated at 15.2 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajkot district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.4% (High fertility class). Soil moisture content is estimated at 26.2%. Soil pH is estimated at 7.41 (on a scale of 0-14). Available nitrogen is estimated at 160 kg/ha, and available phosphorus is estimated at 8.4 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sabar Kantha district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 58.7% (High fertility class). Soil moisture content is estimated at 20.3%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 244 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Surat district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 68.5% (High fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 8.16 (on a scale of 0-14). Available nitrogen is estimated at 227 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot

summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Surendranagar district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 53.6% (Medium fertility class). Soil moisture content is estimated at 30.4%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 19.9 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tapi district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.1% (Medium fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 8.29 (on a scale of 0-14). Available nitrogen is estimated at 226 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

The Dangs district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 72.4% (High fertility class). Soil moisture content is estimated at 33.2%. Soil pH is estimated at 7.7 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Vadodara district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.3% (Medium fertility class). Soil moisture content is estimated at 28.9%. Soil pH is estimated at 7.38 (on a scale of 0-14). Available nitrogen is estimated at 236 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Valsad district, Gujarat. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.5% (Medium fertility class). Soil moisture content is estimated at 28.9%. Soil pH is estimated at 7.3 (on a scale

of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 11.0 kg/ha. The district falls in the Semi-Arid to Arid (Koppen BSh/BWh) climate zone, with typical annual rainfall of 300-2000 mm and an average temperature range of 7 to 43 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep, coastal humidity. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Haryana

Ambala district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.3% (High fertility class). Soil moisture content is estimated at 25.7%. Soil pH is estimated at 7.02 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 23.6 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhiwani district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.2% (High fertility class). Soil moisture content is estimated at 28.6%. Soil pH is estimated at 7.78 (on a scale of 0-14). Available nitrogen is estimated at 238 kg/ha, and available phosphorus is estimated at 20.2 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Faridabad district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 55.3% (High fertility class). Soil moisture content is estimated at 29.1%. Soil pH is estimated at 7.18 (on a scale of 0-14). Available nitrogen is estimated at 237 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Fatehabad district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.2% (High fertility class). Soil moisture content is estimated at 23.5%. Soil pH is estimated at 7.39 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 27.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gurgaon district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.5% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 7.62 (on a scale of 0-14). Available nitrogen is estimated at 307 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hisar district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.7% (High fertility class). Soil moisture content is estimated at 28.3%. Soil pH is estimated at 7.64 (on a scale of 0-14). Available nitrogen is estimated at 233 kg/ha, and available phosphorus is estimated at 20.3 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jhajjar district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 28.2%. Soil pH is estimated at 7.52 (on a scale of 0-14). Available nitrogen is estimated at 259 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jind district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.6% (High fertility class). Soil moisture content is estimated at 30.1%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 193 kg/ha, and available phosphorus is estimated at 15.3 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kaithal district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.6% (High fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 7.77 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 23.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karnal district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.2% (High fertility class). Soil moisture content is estimated at 23.5%. Soil pH is estimated at 6.91 (on a scale of 0-14).

Available nitrogen is estimated at 265 kg/ha, and available phosphorus is estimated at 21.3 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kurukshetra district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.2% (High fertility class). Soil moisture content is estimated at 26.4%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 245 kg/ha, and available phosphorus is estimated at 15.1 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahendragarh district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.4% (High fertility class). Soil moisture content is estimated at 23.5%. Soil pH is estimated at 6.67 (on a scale of 0-14). Available nitrogen is estimated at 202 kg/ha, and available phosphorus is estimated at 23.0 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mewat district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.5% (High fertility class). Soil moisture content is estimated at 31.4%. Soil pH is estimated at 6.82 (on a scale of 0-14). Available nitrogen is estimated at 193 kg/ha, and available phosphorus is estimated at 24.1 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Palwal district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 77.0% (High fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 6.76 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 20.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Panchkula district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.7% (High fertility class). Soil moisture content is estimated at 21.6%. Soil pH is estimated at 6.56 (on a scale of 0-14). Available nitrogen is estimated at 247 kg/ha, and available phosphorus is estimated at 27.3 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of

350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Panipat district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 24.9%. Soil pH is estimated at 6.64 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 16.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rewari district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.1% (High fertility class). Soil moisture content is estimated at 30.3%. Soil pH is estimated at 7.67 (on a scale of 0-14). Available nitrogen is estimated at 317 kg/ha, and available phosphorus is estimated at 27.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rohtak district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.8% (High fertility class). Soil moisture content is estimated at 19.0%. Soil pH is estimated at 6.68 (on a scale of 0-14). Available nitrogen is estimated at 222 kg/ha, and available phosphorus is estimated at 20.8 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sirsa district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.4% (High fertility class). Soil moisture content is estimated at 29.8%. Soil pH is estimated at 7.3 (on a scale of 0-14). Available nitrogen is estimated at 255 kg/ha, and available phosphorus is estimated at 14.6 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sonipat district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.3% (High fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 7.61 (on a scale of 0-14). Available nitrogen is estimated at 261 kg/ha, and available phosphorus is estimated at 23.8 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination

include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Yamunanagar district, Haryana. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.4% (High fertility class). Soil moisture content is estimated at 23.2%. Soil pH is estimated at 7.03 (on a scale of 0-14). Available nitrogen is estimated at 286 kg/ha, and available phosphorus is estimated at 24.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 350-800 mm and an average temperature range of 4 to 44 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Himachal Pradesh

Bilaspur district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 45.2% (Medium fertility class). Soil moisture content is estimated at 37.1%. Soil pH is estimated at 5.22 (on a scale of 0-14). Available nitrogen is estimated at 227 kg/ha, and available phosphorus is estimated at 7.0 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chamba district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.6% (Medium fertility class). Soil moisture content is estimated at 35.5%. Soil pH is estimated at 5.32 (on a scale of 0-14). Available nitrogen is estimated at 169 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hamirpur district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.1% (Medium fertility class). Soil moisture content is estimated at 33.4%. Soil pH is estimated at 6.53 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kangra district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 53.3% (Medium fertility class). Soil moisture content is estimated at 26.0%. Soil pH is estimated at 5.44 (on a scale of 0-14). Available nitrogen is estimated at 139 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius.

Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kinnaur district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 62.2% (High fertility class). Soil moisture content is estimated at 34.4%. Soil pH is estimated at 5.93 (on a scale of 0-14). Available nitrogen is estimated at 178 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kullu district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 53.4% (Medium fertility class). Soil moisture content is estimated at 29.0%. Soil pH is estimated at 5.24 (on a scale of 0-14). Available nitrogen is estimated at 135 kg/ha, and available phosphorus is estimated at 15.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lahul And Spiti district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 52.1% (Medium fertility class). Soil moisture content is estimated at 37.3%. Soil pH is estimated at 5.29 (on a scale of 0-14). Available nitrogen is estimated at 186 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mandi district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.3% (Medium fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 5.21 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 9.2 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shimla district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 55.2% (High fertility class). Soil moisture content is estimated at 29.4%. Soil pH is estimated at 5.72 (on a scale of 0-14). Available nitrogen is estimated at 183 kg/ha, and available phosphorus is estimated at 7.9 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat.

[Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sirmaur district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 64.7% (High fertility class). Soil moisture content is estimated at 35.1%. Soil pH is estimated at 6.33 (on a scale of 0-14). Available nitrogen is estimated at 176 kg/ha, and available phosphorus is estimated at 7.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Solan district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 63.0% (High fertility class). Soil moisture content is estimated at 27.5%. Soil pH is estimated at 5.09 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Una district, Himachal Pradesh. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 34.0%. Soil pH is estimated at 6.43 (on a scale of 0-14). Available nitrogen is estimated at 145 kg/ha, and available phosphorus is estimated at 15.2 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwb) climate zone, with typical annual rainfall of 1000-1800 mm and an average temperature range of (-2) to 32 degrees Celsius. Seasonal pattern: Cold winters, mild-to-warm summers, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jammu And Kashmir

Anantnag district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 50.9% (Medium fertility class). Soil moisture content is estimated at 32.2%. Soil pH is estimated at 5.0 (on a scale of 0-14). Available nitrogen is estimated at 239 kg/ha, and available phosphorus is estimated at 8.8 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Badgam district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 41.6% (Medium fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 5.23 (on a scale of 0-14). Available nitrogen is estimated at 138 kg/ha, and available phosphorus is estimated at 12.3 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically

suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bandipore district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 49.8% (Medium fertility class). Soil moisture content is estimated at 36.7%. Soil pH is estimated at 5.57 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Baramula district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.5% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 5.85 (on a scale of 0-14). Available nitrogen is estimated at 234 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Doda district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 44.4% (Medium fertility class). Soil moisture content is estimated at 32.8%. Soil pH is estimated at 6.52 (on a scale of 0-14). Available nitrogen is estimated at 124 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ganderbal district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 62.3% (High fertility class). Soil moisture content is estimated at 34.0%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 183 kg/ha, and available phosphorus is estimated at 12.6 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jammu district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 47.6% (Medium fertility class). Soil moisture content is estimated at 37.2%. Soil pH is estimated at 5.66 (on a scale of 0-14). Available nitrogen is estimated at 208 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

climatology; not an individual lab measurement or live weather reading.]

Kargil district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 54.0% (Medium fertility class). Soil moisture content is estimated at 33.5%. Soil pH is estimated at 5.92 (on a scale of 0-14). Available nitrogen is estimated at 228 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kathua district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.2% (High fertility class). Soil moisture content is estimated at 27.5%. Soil pH is estimated at 5.73 (on a scale of 0-14). Available nitrogen is estimated at 172 kg/ha, and available phosphorus is estimated at 7.3 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kishtwar district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 42.3% (Medium fertility class). Soil moisture content is estimated at 32.7%. Soil pH is estimated at 6.18 (on a scale of 0-14). Available nitrogen is estimated at 199 kg/ha, and available phosphorus is estimated at 6.2 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kulgam district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 48.2% (Medium fertility class). Soil moisture content is estimated at 25.2%. Soil pH is estimated at 6.4 (on a scale of 0-14). Available nitrogen is estimated at 223 kg/ha, and available phosphorus is estimated at 8.5 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kupwara district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.7% (High fertility class). Soil moisture content is estimated at 24.1%. Soil pH is estimated at 5.5 (on a scale of 0-14). Available nitrogen is estimated at 136 kg/ha, and available phosphorus is estimated at 8.8 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Leh(Ladakh) district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 56.9% (High fertility class). Soil moisture content is estimated at 36.6%. Soil pH is estimated at 5.36 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 13.0 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pulwama district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 63.8% (High fertility class). Soil moisture content is estimated at 31.8%. Soil pH is estimated at 5.46 (on a scale of 0-14). Available nitrogen is estimated at 195 kg/ha, and available phosphorus is estimated at 9.2 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Punch district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 64.4% (High fertility class). Soil moisture content is estimated at 28.4%. Soil pH is estimated at 6.23 (on a scale of 0-14). Available nitrogen is estimated at 122 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajouri district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 61.9% (High fertility class). Soil moisture content is estimated at 31.8%. Soil pH is estimated at 5.8 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 14.5 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ramban district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 48.1% (Medium fertility class). Soil moisture content is estimated at 30.7%. Soil pH is estimated at 5.88 (on a scale of 0-14). Available nitrogen is estimated at 195 kg/ha, and available phosphorus is estimated at 9.3 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Reasi district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.1% (Medium fertility class). Soil moisture content is estimated at 36.6%. Soil pH is

estimated at 6.66 (on a scale of 0-14). Available nitrogen is estimated at 221 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Samba district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 58.6% (High fertility class). Soil moisture content is estimated at 34.2%. Soil pH is estimated at 6.37 (on a scale of 0-14). Available nitrogen is estimated at 177 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shupiyan district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 46.0% (Medium fertility class). Soil moisture content is estimated at 32.3%. Soil pH is estimated at 5.52 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 11.9 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Srinagar district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 50.2% (Medium fertility class). Soil moisture content is estimated at 25.5%. Soil pH is estimated at 6.47 (on a scale of 0-14). Available nitrogen is estimated at 223 kg/ha, and available phosphorus is estimated at 7.7 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Udhampur district, Jammu And Kashmir. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 40.2% (Medium fertility class). Soil moisture content is estimated at 27.4%. Soil pH is estimated at 6.44 (on a scale of 0-14). Available nitrogen is estimated at 220 kg/ha, and available phosphorus is estimated at 7.4 kg/ha. The district falls in the Cold / Alpine (Koppen Dfb/ET) climate zone, with typical annual rainfall of 650-1200 mm and an average temperature range of (-5) to 30 degrees Celsius. Seasonal pattern: Cold winters with snow (Dec-Feb), mild summers (Jun-Aug). Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jharkhand

Bokaro district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 55.0% (High fertility class). Soil moisture content is estimated at 13.2%. Soil pH is estimated at 6.01 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 7.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chatra district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 41.4% (Medium fertility class). Soil moisture content is estimated at 16.2%. Soil pH is estimated at 6.31 (on a scale of 0-14). Available nitrogen is estimated at 212 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Deoghar district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 39.5% (Medium fertility class). Soil moisture content is estimated at 13.4%. Soil pH is estimated at 6.53 (on a scale of 0-14). Available nitrogen is estimated at 139 kg/ha, and available phosphorus is estimated at 14.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhanbad district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 52.6% (Medium fertility class). Soil moisture content is estimated at 13.1%. Soil pH is estimated at 6.58 (on a scale of 0-14). Available nitrogen is estimated at 129 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dumka district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 50.1% (Medium fertility class). Soil moisture content is estimated at 17.0%. Soil pH is estimated at 6.39 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 15.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Garhwa district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 60.0% (High fertility class). Soil moisture content is estimated at 17.2%. Soil pH is estimated at 5.62 (on a scale of 0-14). Available nitrogen is estimated at 190 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red &

Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Giridih district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 45.0% (Medium fertility class). Soil moisture content is estimated at 16.6%. Soil pH is estimated at 6.05 (on a scale of 0-14). Available nitrogen is estimated at 115 kg/ha, and available phosphorus is estimated at 12.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Godda district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 45.8% (Medium fertility class). Soil moisture content is estimated at 19.9%. Soil pH is estimated at 6.58 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gumla district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 45.1% (Medium fertility class). Soil moisture content is estimated at 19.5%. Soil pH is estimated at 6.54 (on a scale of 0-14). Available nitrogen is estimated at 164 kg/ha, and available phosphorus is estimated at 14.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hazaribagh district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 41.2% (Medium fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 5.62 (on a scale of 0-14). Available nitrogen is estimated at 126 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jamtara district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 39.5% (Medium fertility class). Soil moisture content is estimated at 12.5%. Soil pH is estimated at 5.5 (on a scale of 0-14). Available nitrogen is estimated at 157 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khunti district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 38.8% (Medium fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 7.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot

summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kodarma district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 56.9% (High fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 6.2 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 6.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Latehar district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 46.7% (Medium fertility class). Soil moisture content is estimated at 19.4%. Soil pH is estimated at 5.83 (on a scale of 0-14). Available nitrogen is estimated at 192 kg/ha, and available phosphorus is estimated at 11.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lohardaga district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 54.5% (Medium fertility class). Soil moisture content is estimated at 16.6%. Soil pH is estimated at 5.98 (on a scale of 0-14). Available nitrogen is estimated at 134 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pakur district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 54.5% (Medium fertility class). Soil moisture content is estimated at 13.2%. Soil pH is estimated at 5.61 (on a scale of 0-14). Available nitrogen is estimated at 129 kg/ha, and available phosphorus is estimated at 7.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Palamu district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 16.0%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 125 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pashchimi Singhbhum district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 47.9% (Medium fertility class). Soil moisture content is estimated at 15.9%. Soil pH is estimated at 6.64 (on a scale of 0-14). Available nitrogen is estimated at 217 kg/ha, and available

phosphorus is estimated at 14.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Purbi Singhbhum district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 42.2% (Medium fertility class). Soil moisture content is estimated at 14.2%. Soil pH is estimated at 6.35 (on a scale of 0-14). Available nitrogen is estimated at 141 kg/ha, and available phosphorus is estimated at 13.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ramgarh district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 50.5% (Medium fertility class). Soil moisture content is estimated at 17.5%. Soil pH is estimated at 6.58 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 6.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ranchi district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 39.1% (Medium fertility class). Soil moisture content is estimated at 18.0%. Soil pH is estimated at 6.13 (on a scale of 0-14). Available nitrogen is estimated at 138 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sahibganj district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 54.7% (Medium fertility class). Soil moisture content is estimated at 20.3%. Soil pH is estimated at 6.23 (on a scale of 0-14). Available nitrogen is estimated at 152 kg/ha, and available phosphorus is estimated at 8.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Saraikela-Kharsawan district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 49.5% (Medium fertility class). Soil moisture content is estimated at 15.1%. Soil pH is estimated at 6.23 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 6.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Simdega district, Jharkhand. Dominant soil type: Red & Yellow Soil. Soil fertility is estimated at 38.0% (Medium fertility class). Soil moisture content is estimated at 14.1%. Soil pH is estimated at 6.66 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 7.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1100-1500 mm and an average temperature range of 6 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets (ragi, bajra), pulses, groundnut and potato. [Basis: indicative regional estimate for Red & Yellow Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karnataka

Bagalkot district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.5% (High fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 6.82 (on a scale of 0-14). Available nitrogen is estimated at 242 kg/ha, and available phosphorus is estimated at 9.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bangalore district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.9% (Medium fertility class). Soil moisture content is estimated at 21.0%. Soil pH is estimated at 6.45 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 11.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bangalore Rural district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 47.3% (Medium fertility class). Soil moisture content is estimated at 16.5%. Soil pH is estimated at 6.75 (on a scale of 0-14). Available nitrogen is estimated at 159 kg/ha, and available phosphorus is estimated at 10.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Belgaum district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.8% (High fertility class). Soil moisture content is estimated at 17.8%. Soil pH is estimated at 6.67 (on a scale of 0-14). Available nitrogen is estimated at 146 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bellary district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 51.1% (Medium fertility class). Soil moisture content is estimated at 21.3%. Soil pH is estimated at 6.47 (on a scale of 0-14). Available nitrogen is estimated at 237 kg/ha, and available phosphorus is estimated at 21.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bidar district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 47.3% (Medium fertility class). Soil moisture content is estimated at 15.3%. Soil pH is estimated at 6.35 (on a scale of 0-14). Available nitrogen is estimated at 208 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bijapur district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.6% (High fertility class). Soil moisture content is estimated at 23.1%. Soil pH is estimated at 7.27 (on a scale of 0-14). Available nitrogen is estimated at 197 kg/ha, and available phosphorus is estimated at 22.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chamarajanagar district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 46.0% (Medium fertility class). Soil moisture content is estimated at 18.8%. Soil pH is estimated at 6.62 (on a scale of 0-14). Available nitrogen is estimated at 244 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chikkaballapura district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 63.5% (High fertility class). Soil moisture content is estimated at 16.0%. Soil pH is estimated at 7.47 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chikmagalur district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 67.3% (High fertility class). Soil moisture content is estimated at 18.6%. Soil pH is estimated at 6.72 (on a scale of

0-14). Available nitrogen is estimated at 204 kg/ha, and available phosphorus is estimated at 10.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chitradurga district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 52.4% (Medium fertility class). Soil moisture content is estimated at 16.3%. Soil pH is estimated at 6.33 (on a scale of 0-14). Available nitrogen is estimated at 235 kg/ha, and available phosphorus is estimated at 16.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dakshina Kannada district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 53.2% (Medium fertility class). Soil moisture content is estimated at 18.9%. Soil pH is estimated at 6.76 (on a scale of 0-14). Available nitrogen is estimated at 209 kg/ha, and available phosphorus is estimated at 21.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Davanagere district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 66.5% (High fertility class). Soil moisture content is estimated at 18.7%. Soil pH is estimated at 6.63 (on a scale of 0-14). Available nitrogen is estimated at 189 kg/ha, and available phosphorus is estimated at 16.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dharwad district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 6.92 (on a scale of 0-14). Available nitrogen is estimated at 175 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gadag district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 53.7% (Medium fertility class). Soil moisture content is estimated at 15.1%. Soil pH is estimated at 6.42 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 14.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall

of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gulbarga district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 49.9% (Medium fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 7.18 (on a scale of 0-14). Available nitrogen is estimated at 168 kg/ha, and available phosphorus is estimated at 19.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hassan district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 66.3% (High fertility class). Soil moisture content is estimated at 21.2%. Soil pH is estimated at 7.07 (on a scale of 0-14). Available nitrogen is estimated at 145 kg/ha, and available phosphorus is estimated at 21.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Haveri district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 54.0% (Medium fertility class). Soil moisture content is estimated at 25.1%. Soil pH is estimated at 6.95 (on a scale of 0-14). Available nitrogen is estimated at 140 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kodagu district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 60.6% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 6.23 (on a scale of 0-14). Available nitrogen is estimated at 221 kg/ha, and available phosphorus is estimated at 20.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kolar district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 6.21 (on a scale of 0-14). Available nitrogen is estimated at 226 kg/ha, and available phosphorus is estimated at 13.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate

combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Koppal district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.2% (High fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 6.93 (on a scale of 0-14). Available nitrogen is estimated at 169 kg/ha, and available phosphorus is estimated at 11.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mandya district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 62.8% (High fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 6.9 (on a scale of 0-14). Available nitrogen is estimated at 246 kg/ha, and available phosphorus is estimated at 21.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mysore district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.7% (High fertility class). Soil moisture content is estimated at 16.3%. Soil pH is estimated at 6.97 (on a scale of 0-14). Available nitrogen is estimated at 236 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Raichur district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 47.3% (Medium fertility class). Soil moisture content is estimated at 23.7%. Soil pH is estimated at 6.83 (on a scale of 0-14). Available nitrogen is estimated at 202 kg/ha, and available phosphorus is estimated at 21.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ramanagara district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 59.0% (High fertility class). Soil moisture content is estimated at 19.1%. Soil pH is estimated at 6.68 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

weather reading.]

Shimoga district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 63.3% (High fertility class). Soil moisture content is estimated at 24.1%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 173 kg/ha, and available phosphorus is estimated at 20.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tumkur district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 45.6% (Medium fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 6.8 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 11.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Udupi district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 60.4% (High fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 7.0 (on a scale of 0-14). Available nitrogen is estimated at 144 kg/ha, and available phosphorus is estimated at 15.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttara Kannada district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 57.0% (High fertility class). Soil moisture content is estimated at 16.6%. Soil pH is estimated at 6.38 (on a scale of 0-14). Available nitrogen is estimated at 160 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Yadgir district, Karnataka. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 47.2% (Medium fertility class). Soil moisture content is estimated at 25.4%. Soil pH is estimated at 6.73 (on a scale of 0-14). Available nitrogen is estimated at 228 kg/ha, and available phosphorus is estimated at 21.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Am) climate zone, with typical annual rainfall of 600-3800 mm and an average temperature range of 11 to 38 degrees Celsius. Seasonal pattern: Coastal humidity, drier interior plateau, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kerala

Alappuzha district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 33.9% (Low fertility class). Soil moisture content is estimated at 29.8%. Soil pH is estimated at 5.63 (on a scale of 0-14). Available nitrogen is estimated at 153 kg/ha, and available phosphorus is estimated at 7.4 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ernakulam district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 41.9% (Medium fertility class). Soil moisture content is estimated at 20.5%. Soil pH is estimated at 5.19 (on a scale of 0-14). Available nitrogen is estimated at 189 kg/ha, and available phosphorus is estimated at 11.7 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Idukki district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 32.9% (Low fertility class). Soil moisture content is estimated at 30.6%. Soil pH is estimated at 4.89 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 4.4 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kannur district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 53.5% (Medium fertility class). Soil moisture content is estimated at 32.6%. Soil pH is estimated at 4.65 (on a scale of 0-14). Available nitrogen is estimated at 95 kg/ha, and available phosphorus is estimated at 10.7 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kasaragod district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 35.6% (Medium fertility class). Soil moisture content is estimated at 28.0%. Soil pH is estimated at 5.69 (on a scale of 0-14). Available nitrogen is estimated at 146 kg/ha, and available phosphorus is estimated at 7.2 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kollam district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 42.2% (Medium fertility class). Soil moisture content is estimated at 20.9%. Soil pH is estimated at 5.76 (on a scale of 0-14). Available nitrogen is estimated at 121 kg/ha, and available phosphorus is estimated at 5.5 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kottayam district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 38.6% (Medium fertility class). Soil moisture content is estimated at 34.5%. Soil pH is estimated at 4.62 (on a scale of 0-14). Available nitrogen is estimated at 176 kg/ha, and available phosphorus is estimated at 9.7 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kozhikode district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 51.1% (Medium fertility class). Soil moisture content is estimated at 34.6%. Soil pH is estimated at 5.59 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 5.0 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Malappuram district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 50.3% (Medium fertility class). Soil moisture content is estimated at 28.2%. Soil pH is estimated at 5.21 (on a scale of 0-14). Available nitrogen is estimated at 179 kg/ha, and available phosphorus is estimated at 11.8 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Palakkad district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 48.2% (Medium fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 5.09 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 6.9 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pathanamthitta district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 41.0% (Medium fertility class). Soil moisture content is estimated at 21.3%. Soil pH is estimated at 4.6 (on a scale

of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 7.6 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thiruvananthapuram district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 53.4% (Medium fertility class). Soil moisture content is estimated at 29.2%. Soil pH is estimated at 4.55 (on a scale of 0-14). Available nitrogen is estimated at 105 kg/ha, and available phosphorus is estimated at 5.6 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thrissur district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 54.4% (Medium fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 5.17 (on a scale of 0-14). Available nitrogen is estimated at 183 kg/ha, and available phosphorus is estimated at 11.8 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Wayanad district, Kerala. Dominant soil type: Laterite Soil. Soil fertility is estimated at 37.4% (Medium fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 4.7 (on a scale of 0-14). Available nitrogen is estimated at 96 kg/ha, and available phosphorus is estimated at 5.6 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 2500-3500 mm and an average temperature range of 21 to 35 degrees Celsius. Seasonal pattern: Hot humid, two monsoons (Jun-Sep, Oct-Nov). Crops typically suited to this soil and climate combination include: rice, cashew, coconut, rubber, tea and spices; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lakshadweep

Lakshadweep district, Lakshadweep. Dominant soil type: Coastal Sandy Soil. Soil fertility is estimated at 34.7% (Low fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 7.2 (on a scale of 0-14). Available nitrogen is estimated at 110 kg/ha, and available phosphorus is estimated at 5.2 kg/ha. The district falls in the Tropical Monsoon (Koppen Am) climate zone, with typical annual rainfall of 1600-1800 mm and an average temperature range of 22 to 32 degrees Celsius. Seasonal pattern: Warm humid maritime climate year-round. Crops typically suited to this soil and climate combination include: coconut and fisheries-linked horticulture. [Basis: indicative regional estimate for Coastal Sandy Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Madhya Pradesh

Alirajpur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 70.0% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 8.12 (on a scale of 0-14). Available nitrogen is estimated at 273 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Anuppur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 52.3% (Medium fertility class). Soil moisture content is estimated at 23.4%. Soil pH is estimated at 7.57 (on a scale of 0-14). Available nitrogen is estimated at 272 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ashoknagar district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.3% (High fertility class). Soil moisture content is estimated at 28.1%. Soil pH is estimated at 7.24 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Balaghat district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 66.8% (High fertility class). Soil moisture content is estimated at 26.5%. Soil pH is estimated at 7.3 (on a scale of 0-14). Available nitrogen is estimated at 243 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Barwani district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 73.4% (High fertility class). Soil moisture content is estimated at 22.8%. Soil pH is estimated at 7.14 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 14.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Betul district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.7% (High fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 8.29 (on a

scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhind district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 60.5% (High fertility class). Soil moisture content is estimated at 21.0%. Soil pH is estimated at 7.77 (on a scale of 0-14). Available nitrogen is estimated at 179 kg/ha, and available phosphorus is estimated at 15.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhopal district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.3% (Medium fertility class). Soil moisture content is estimated at 34.7%. Soil pH is estimated at 7.61 (on a scale of 0-14). Available nitrogen is estimated at 266 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Burhanpur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.5% (Medium fertility class). Soil moisture content is estimated at 32.3%. Soil pH is estimated at 7.84 (on a scale of 0-14). Available nitrogen is estimated at 251 kg/ha, and available phosphorus is estimated at 13.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chhatarpur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 65.9% (High fertility class). Soil moisture content is estimated at 32.8%. Soil pH is estimated at 7.82 (on a scale of 0-14). Available nitrogen is estimated at 195 kg/ha, and available phosphorus is estimated at 16.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chhindwara district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 66.8% (High fertility class). Soil moisture content is estimated at 34.4%. Soil pH is estimated at 7.48 (on a scale of 0-14). Available nitrogen is estimated at 205 kg/ha, and available phosphorus is estimated at 17.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual

rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Damoh district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.6% (High fertility class). Soil moisture content is estimated at 31.0%. Soil pH is estimated at 7.6 (on a scale of 0-14). Available nitrogen is estimated at 178 kg/ha, and available phosphorus is estimated at 16.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Datia district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 53.3% (Medium fertility class). Soil moisture content is estimated at 23.0%. Soil pH is estimated at 7.81 (on a scale of 0-14). Available nitrogen is estimated at 247 kg/ha, and available phosphorus is estimated at 18.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dewas district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.8% (High fertility class). Soil moisture content is estimated at 21.7%. Soil pH is estimated at 8.24 (on a scale of 0-14). Available nitrogen is estimated at 170 kg/ha, and available phosphorus is estimated at 10.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhar district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.3% (High fertility class). Soil moisture content is estimated at 27.9%. Soil pH is estimated at 8.15 (on a scale of 0-14). Available nitrogen is estimated at 248 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dindori district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 54.4% (Medium fertility class). Soil moisture content is estimated at 25.4%. Soil pH is estimated at 7.62 (on a scale of 0-14). Available nitrogen is estimated at 169 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate

combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Guna district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.7% (High fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 241 kg/ha, and available phosphorus is estimated at 8.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gwalior district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.0% (High fertility class). Soil moisture content is estimated at 31.7%. Soil pH is estimated at 7.97 (on a scale of 0-14). Available nitrogen is estimated at 255 kg/ha, and available phosphorus is estimated at 13.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Harda district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 70.5% (High fertility class). Soil moisture content is estimated at 23.6%. Soil pH is estimated at 7.47 (on a scale of 0-14). Available nitrogen is estimated at 172 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hoshangabad district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 59.8% (High fertility class). Soil moisture content is estimated at 23.6%. Soil pH is estimated at 7.98 (on a scale of 0-14). Available nitrogen is estimated at 190 kg/ha, and available phosphorus is estimated at 11.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Indore district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 53.3% (Medium fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 7.39 (on a scale of 0-14). Available nitrogen is estimated at 198 kg/ha, and available phosphorus is estimated at 10.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement

or live weather reading.]

Jabalpur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.1% (High fertility class). Soil moisture content is estimated at 34.0%. Soil pH is estimated at 7.65 (on a scale of 0-14). Available nitrogen is estimated at 233 kg/ha, and available phosphorus is estimated at 18.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jhabua district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 75.0% (High fertility class). Soil moisture content is estimated at 29.2%. Soil pH is estimated at 8.16 (on a scale of 0-14). Available nitrogen is estimated at 268 kg/ha, and available phosphorus is estimated at 17.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Katni district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 61.5% (High fertility class). Soil moisture content is estimated at 21.2%. Soil pH is estimated at 7.02 (on a scale of 0-14). Available nitrogen is estimated at 212 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khandwa (East Nimar) district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 60.2% (High fertility class). Soil moisture content is estimated at 28.5%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 258 kg/ha, and available phosphorus is estimated at 9.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khargone (West Nimar) district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 31.9%. Soil pH is estimated at 8.1 (on a scale of 0-14). Available nitrogen is estimated at 279 kg/ha, and available phosphorus is estimated at 18.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mandla district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 72.2% (High fertility class). Soil moisture content is estimated at 24.7%. Soil pH is estimated at 7.52 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 15.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mandsaur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 25.2%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Morena district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 54.9% (Medium fertility class). Soil moisture content is estimated at 20.9%. Soil pH is estimated at 8.23 (on a scale of 0-14). Available nitrogen is estimated at 202 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Narsimhapur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 67.0% (High fertility class). Soil moisture content is estimated at 22.9%. Soil pH is estimated at 7.23 (on a scale of 0-14). Available nitrogen is estimated at 255 kg/ha, and available phosphorus is estimated at 11.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Neemuch district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 72.8% (High fertility class). Soil moisture content is estimated at 22.1%. Soil pH is estimated at 7.88 (on a scale of 0-14). Available nitrogen is estimated at 255 kg/ha, and available phosphorus is estimated at 10.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Panna district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.9% (High fertility class). Soil moisture content is estimated at 34.5%. Soil pH is estimated at 7.34 (on a

scale of 0-14). Available nitrogen is estimated at 205 kg/ha, and available phosphorus is estimated at 13.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Raisen district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.3% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 8.0 (on a scale of 0-14). Available nitrogen is estimated at 199 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajgarh district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.3% (High fertility class). Soil moisture content is estimated at 20.1%. Soil pH is estimated at 7.74 (on a scale of 0-14). Available nitrogen is estimated at 279 kg/ha, and available phosphorus is estimated at 8.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ratlam district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.2% (High fertility class). Soil moisture content is estimated at 27.8%. Soil pH is estimated at 8.05 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 12.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rewa district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 68.7% (High fertility class). Soil moisture content is estimated at 25.6%. Soil pH is estimated at 7.25 (on a scale of 0-14). Available nitrogen is estimated at 175 kg/ha, and available phosphorus is estimated at 12.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sagar district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 62.0% (High fertility class). Soil moisture content is estimated at 32.9%. Soil pH is estimated at 7.83 (on a scale of 0-14). Available nitrogen is estimated at 239 kg/ha, and available phosphorus is estimated at 19.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of

800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Satna district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 60.3% (High fertility class). Soil moisture content is estimated at 26.1%. Soil pH is estimated at 8.2 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 16.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sehore district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 68.9% (High fertility class). Soil moisture content is estimated at 21.5%. Soil pH is estimated at 7.61 (on a scale of 0-14). Available nitrogen is estimated at 176 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Seoni district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 61.6% (High fertility class). Soil moisture content is estimated at 24.3%. Soil pH is estimated at 8.03 (on a scale of 0-14). Available nitrogen is estimated at 267 kg/ha, and available phosphorus is estimated at 17.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shahdol district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.7% (High fertility class). Soil moisture content is estimated at 24.1%. Soil pH is estimated at 7.33 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 19.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shajapur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 60.1% (High fertility class). Soil moisture content is estimated at 23.4%. Soil pH is estimated at 8.1 (on a scale of 0-14). Available nitrogen is estimated at 201 kg/ha, and available phosphorus is estimated at 8.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination

include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sheopur district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 54.0% (Medium fertility class). Soil moisture content is estimated at 29.6%. Soil pH is estimated at 8.02 (on a scale of 0-14). Available nitrogen is estimated at 177 kg/ha, and available phosphorus is estimated at 19.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shivpuri district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 32.7%. Soil pH is estimated at 7.3 (on a scale of 0-14). Available nitrogen is estimated at 156 kg/ha, and available phosphorus is estimated at 18.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sidhi district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.6% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 7.99 (on a scale of 0-14). Available nitrogen is estimated at 166 kg/ha, and available phosphorus is estimated at 17.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Singrauli district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 30.9%. Soil pH is estimated at 7.57 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 14.3 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tikamgarh district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 68.6% (High fertility class). Soil moisture content is estimated at 32.3%. Soil pH is estimated at 7.97 (on a scale of 0-14). Available nitrogen is estimated at 187 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

weather reading.]

Ujjain district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.0% (High fertility class). Soil moisture content is estimated at 27.5%. Soil pH is estimated at 7.66 (on a scale of 0-14). Available nitrogen is estimated at 225 kg/ha, and available phosphorus is estimated at 12.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Umari district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.7% (High fertility class). Soil moisture content is estimated at 33.7%. Soil pH is estimated at 7.32 (on a scale of 0-14). Available nitrogen is estimated at 245 kg/ha, and available phosphorus is estimated at 16.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Vidisha district, Madhya Pradesh. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 63.1% (High fertility class). Soil moisture content is estimated at 22.8%. Soil pH is estimated at 8.03 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 16.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 800-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot dry summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Maharashtra

Ahmadnagar district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 72.9% (High fertility class). Soil moisture content is estimated at 34.8%. Soil pH is estimated at 7.46 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Akola district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 74.7% (High fertility class). Soil moisture content is estimated at 21.4%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 11.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative

regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Amravati district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.7% (High fertility class). Soil moisture content is estimated at 25.7%. Soil pH is estimated at 8.27 (on a scale of 0-14). Available nitrogen is estimated at 264 kg/ha, and available phosphorus is estimated at 11.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Aurangabad district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.3% (High fertility class). Soil moisture content is estimated at 25.1%. Soil pH is estimated at 7.79 (on a scale of 0-14). Available nitrogen is estimated at 235 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhandara district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.6% (Medium fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 7.93 (on a scale of 0-14). Available nitrogen is estimated at 223 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bid district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.4% (Medium fertility class). Soil moisture content is estimated at 25.2%. Soil pH is estimated at 8.12 (on a scale of 0-14). Available nitrogen is estimated at 154 kg/ha, and available phosphorus is estimated at 12.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Buldana district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 73.6% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 7.97 (on a scale of 0-14). Available nitrogen is estimated at 183 kg/ha, and available phosphorus is estimated at 18.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chandrapur district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 68.7% (High fertility class). Soil moisture content is estimated at 33.3%. Soil pH is estimated at 7.28 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 17.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhule district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 23.5%. Soil pH is estimated at 7.04 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gadchiroli district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 65.9% (High fertility class). Soil moisture content is estimated at 31.2%. Soil pH is estimated at 7.69 (on a scale of 0-14). Available nitrogen is estimated at 211 kg/ha, and available phosphorus is estimated at 13.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gondiya district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 66.0% (High fertility class). Soil moisture content is estimated at 23.0%. Soil pH is estimated at 8.15 (on a scale of 0-14). Available nitrogen is estimated at 255 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hingoli district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 64.5% (High fertility class). Soil moisture content is estimated at 22.1%. Soil pH is estimated at 8.26 (on a scale of 0-14). Available nitrogen is estimated at 267 kg/ha, and available phosphorus is estimated at 19.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalgaon district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 58.3% (High fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 7.28 (on a scale of

0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalna district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.8% (Medium fertility class). Soil moisture content is estimated at 34.8%. Soil pH is estimated at 7.1 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 17.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kolhapur district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 64.5% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 7.82 (on a scale of 0-14). Available nitrogen is estimated at 264 kg/ha, and available phosphorus is estimated at 15.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Latur district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 63.1% (High fertility class). Soil moisture content is estimated at 30.7%. Soil pH is estimated at 7.58 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mumbai district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 54.2% (Medium fertility class). Soil moisture content is estimated at 31.9%. Soil pH is estimated at 7.34 (on a scale of 0-14). Available nitrogen is estimated at 169 kg/ha, and available phosphorus is estimated at 11.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mumbai Suburban district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 71.1% (High fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 7.89 (on a scale of 0-14). Available nitrogen is estimated at 261 kg/ha, and available phosphorus is estimated at 8.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical

annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nagpur district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 69.1% (High fertility class). Soil moisture content is estimated at 24.6%. Soil pH is estimated at 7.28 (on a scale of 0-14). Available nitrogen is estimated at 239 kg/ha, and available phosphorus is estimated at 18.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nanded district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.4% (High fertility class). Soil moisture content is estimated at 28.5%. Soil pH is estimated at 7.38 (on a scale of 0-14). Available nitrogen is estimated at 222 kg/ha, and available phosphorus is estimated at 15.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nandurbar district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 65.3% (High fertility class). Soil moisture content is estimated at 29.4%. Soil pH is estimated at 7.18 (on a scale of 0-14). Available nitrogen is estimated at 240 kg/ha, and available phosphorus is estimated at 16.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nashik district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 67.1% (High fertility class). Soil moisture content is estimated at 21.1%. Soil pH is estimated at 7.58 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 9.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Osmanabad district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 7.97 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate

combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Parbhani district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 63.6% (High fertility class). Soil moisture content is estimated at 31.4%. Soil pH is estimated at 7.61 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pune district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 73.2% (High fertility class). Soil moisture content is estimated at 27.4%. Soil pH is estimated at 7.69 (on a scale of 0-14). Available nitrogen is estimated at 228 kg/ha, and available phosphorus is estimated at 16.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Raigarh district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 51.8% (Medium fertility class). Soil moisture content is estimated at 27.7%. Soil pH is estimated at 7.63 (on a scale of 0-14). Available nitrogen is estimated at 269 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ratnagiri district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 61.7% (High fertility class). Soil moisture content is estimated at 34.4%. Soil pH is estimated at 7.59 (on a scale of 0-14). Available nitrogen is estimated at 173 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sangli district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 52.4% (Medium fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 7.56 (on a scale of 0-14). Available nitrogen is estimated at 250 kg/ha, and available phosphorus is estimated at 16.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

or live weather reading.]

Satara district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 50.8% (Medium fertility class). Soil moisture content is estimated at 28.5%. Soil pH is estimated at 7.2 (on a scale of 0-14). Available nitrogen is estimated at 272 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sindhudurg district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 65.7% (High fertility class). Soil moisture content is estimated at 24.9%. Soil pH is estimated at 7.83 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 17.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Solapur district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 55.8% (High fertility class). Soil moisture content is estimated at 33.6%. Soil pH is estimated at 7.65 (on a scale of 0-14). Available nitrogen is estimated at 248 kg/ha, and available phosphorus is estimated at 18.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thane district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 75.0% (High fertility class). Soil moisture content is estimated at 23.4%. Soil pH is estimated at 7.37 (on a scale of 0-14). Available nitrogen is estimated at 182 kg/ha, and available phosphorus is estimated at 11.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Wardha district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 64.6% (High fertility class). Soil moisture content is estimated at 34.9%. Soil pH is estimated at 8.29 (on a scale of 0-14). Available nitrogen is estimated at 218 kg/ha, and available phosphorus is estimated at 15.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Washim district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 71.5% (High fertility class). Soil moisture content is estimated at 34.0%. Soil pH is estimated at 7.27 (on a scale of 0-14). Available nitrogen is estimated at 218 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Yavatmal district, Maharashtra. Dominant soil type: Black Soil (Regur). Soil fertility is estimated at 57.6% (High fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 8.28 (on a scale of 0-14). Available nitrogen is estimated at 278 kg/ha, and available phosphorus is estimated at 18.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw) climate zone, with typical annual rainfall of 600-3500 mm and an average temperature range of 9 to 42 degrees Celsius. Seasonal pattern: Hot summers, mild winters, heavy monsoon on Konkan coast. Crops typically suited to this soil and climate combination include: cotton, soybean, sorghum (jowar), sugarcane, pulses and citrus. [Basis: indicative regional estimate for Black Soil (Regur) soil and state-level climatology; not an individual lab measurement or live weather reading.]

Manipur

Bishnupur district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 53.4% (Medium fertility class). Soil moisture content is estimated at 21.8%. Soil pH is estimated at 5.25 (on a scale of 0-14). Available nitrogen is estimated at 154 kg/ha, and available phosphorus is estimated at 10.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chandel district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 46.6% (Medium fertility class). Soil moisture content is estimated at 26.9%. Soil pH is estimated at 5.37 (on a scale of 0-14). Available nitrogen is estimated at 174 kg/ha, and available phosphorus is estimated at 8.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Churachandpur district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 42.6% (Medium fertility class). Soil moisture content is estimated at 19.8%. Soil pH is estimated at 5.06 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 9.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Imphal East district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 40.2% (Medium fertility class). Soil moisture content is estimated at 28.6%. Soil pH is estimated at 5.58 (on a

scale of 0-14). Available nitrogen is estimated at 102 kg/ha, and available phosphorus is estimated at 7.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Imphal West district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 39.1% (Medium fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 5.19 (on a scale of 0-14). Available nitrogen is estimated at 106 kg/ha, and available phosphorus is estimated at 5.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Senapati district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 48.4% (Medium fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 5.99 (on a scale of 0-14). Available nitrogen is estimated at 115 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tamenglong district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 45.2% (Medium fertility class). Soil moisture content is estimated at 24.2%. Soil pH is estimated at 4.9 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 13.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thoubal district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 35.3% (Medium fertility class). Soil moisture content is estimated at 20.5%. Soil pH is estimated at 6.07 (on a scale of 0-14). Available nitrogen is estimated at 154 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ukhrul district, Manipur. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 39.7% (Medium fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 5.81 (on a scale of 0-14). Available nitrogen is estimated at 159 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1400-2500 mm and an average temperature range of 2 to 32 degrees Celsius. Seasonal pattern: Mild summers, cool winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Meghalaya

East Garo Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 46.4% (Medium fertility class). Soil moisture content is estimated at 22.2%. Soil pH is estimated at 5.47 (on a scale of 0-14). Available nitrogen is estimated at 135 kg/ha, and available phosphorus is estimated at 7.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

East Khasi Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 36.1% (Medium fertility class). Soil moisture content is estimated at 22.9%. Soil pH is estimated at 5.8 (on a scale of 0-14). Available nitrogen is estimated at 101 kg/ha, and available phosphorus is estimated at 9.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jaintia Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 49.8% (Medium fertility class). Soil moisture content is estimated at 26.5%. Soil pH is estimated at 5.89 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 11.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ribhoi district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 37.4% (Medium fertility class). Soil moisture content is estimated at 24.7%. Soil pH is estimated at 5.64 (on a scale of 0-14). Available nitrogen is estimated at 124 kg/ha, and available phosphorus is estimated at 5.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South Garo Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 40.1% (Medium fertility class). Soil moisture content is estimated at 24.4%. Soil pH is estimated at 5.38 (on a scale of 0-14). Available nitrogen is estimated at 117 kg/ha, and available phosphorus is estimated at 12.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically

suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Garo Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 56.3% (High fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 6.1 (on a scale of 0-14). Available nitrogen is estimated at 101 kg/ha, and available phosphorus is estimated at 13.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Khasi Hills district, Meghalaya. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 56.9% (High fertility class). Soil moisture content is estimated at 18.2%. Soil pH is estimated at 5.87 (on a scale of 0-14). Available nitrogen is estimated at 197 kg/ha, and available phosphorus is estimated at 13.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwb) climate zone, with typical annual rainfall of 2500-11000 mm and an average temperature range of 4 to 28 degrees Celsius. Seasonal pattern: Among the wettest regions on Earth (Cherrapunji/Mawsynram), cool year-round. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mizoram

Aizawl district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 57.3% (High fertility class). Soil moisture content is estimated at 25.7%. Soil pH is estimated at 4.91 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Champhai district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 49.1% (Medium fertility class). Soil moisture content is estimated at 23.8%. Soil pH is estimated at 5.6 (on a scale of 0-14). Available nitrogen is estimated at 115 kg/ha, and available phosphorus is estimated at 11.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kolasib district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 36.7% (Medium fertility class). Soil moisture content is estimated at 28.2%. Soil pH is estimated at 5.99 (on a

scale of 0-14). Available nitrogen is estimated at 147 kg/ha, and available phosphorus is estimated at 9.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lawngtlai district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 42.2% (Medium fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 5.83 (on a scale of 0-14). Available nitrogen is estimated at 136 kg/ha, and available phosphorus is estimated at 6.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lunglei district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 40.1% (Medium fertility class). Soil moisture content is estimated at 24.5%. Soil pH is estimated at 6.12 (on a scale of 0-14). Available nitrogen is estimated at 173 kg/ha, and available phosphorus is estimated at 6.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mamit district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 54.8% (Medium fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 5.99 (on a scale of 0-14). Available nitrogen is estimated at 124 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Saiha district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 56.3% (High fertility class). Soil moisture content is estimated at 21.6%. Soil pH is estimated at 5.77 (on a scale of 0-14). Available nitrogen is estimated at 198 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Serchhip district, Mizoram. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 37.0% (Medium fertility class). Soil moisture content is estimated at 27.4%. Soil pH is estimated at 5.28 (on a scale of 0-14). Available nitrogen is estimated at 107 kg/ha, and available phosphorus is estimated at 9.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of

2000-3000 mm and an average temperature range of 7 to 30 degrees Celsius. Seasonal pattern: Mild year-round, heavy monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nagaland

Dimapur district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 52.3% (Medium fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 5.27 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 9.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kiphire district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 54.4% (Medium fertility class). Soil moisture content is estimated at 24.2%. Soil pH is estimated at 6.13 (on a scale of 0-14). Available nitrogen is estimated at 165 kg/ha, and available phosphorus is estimated at 6.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kohima district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 55.4% (High fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 5.0 (on a scale of 0-14). Available nitrogen is estimated at 115 kg/ha, and available phosphorus is estimated at 11.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Longleng district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 36.1% (Medium fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 5.0 (on a scale of 0-14). Available nitrogen is estimated at 144 kg/ha, and available phosphorus is estimated at 8.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mokokchung district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 47.0% (Medium fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 4.84 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 8.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red &

Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mon district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 50.0% (Medium fertility class). Soil moisture content is estimated at 19.5%. Soil pH is estimated at 4.82 (on a scale of 0-14). Available nitrogen is estimated at 124 kg/ha, and available phosphorus is estimated at 9.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Peren district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 44.1% (Medium fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 5.33 (on a scale of 0-14). Available nitrogen is estimated at 153 kg/ha, and available phosphorus is estimated at 10.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Phek district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 35.5% (Medium fertility class). Soil moisture content is estimated at 25.2%. Soil pH is estimated at 5.28 (on a scale of 0-14). Available nitrogen is estimated at 142 kg/ha, and available phosphorus is estimated at 5.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tuensang district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 53.0% (Medium fertility class). Soil moisture content is estimated at 31.2%. Soil pH is estimated at 4.83 (on a scale of 0-14). Available nitrogen is estimated at 160 kg/ha, and available phosphorus is estimated at 11.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Wokha district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 41.6% (Medium fertility class). Soil moisture content is estimated at 29.0%. Soil pH is estimated at 5.69 (on a scale of 0-14). Available nitrogen is estimated at 192 kg/ha, and available phosphorus is estimated at 13.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Zunheboto district, Nagaland. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 36.0% (Medium fertility class). Soil moisture content is estimated at 24.2%. Soil pH is estimated at 5.57 (on a scale of 0-14). Available nitrogen is estimated at 114 kg/ha, and available phosphorus is estimated at 8.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 1800-2500 mm and an average temperature range of 4 to 31 degrees Celsius. Seasonal pattern: Mild

summers, cool winters, heavy monsoon. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Odisha

Anugul district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 57.7% (High fertility class). Soil moisture content is estimated at 15.2%. Soil pH is estimated at 5.68 (on a scale of 0-14). Available nitrogen is estimated at 192 kg/ha, and available phosphorus is estimated at 7.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Balangir district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 46.0% (Medium fertility class). Soil moisture content is estimated at 27.7%. Soil pH is estimated at 5.71 (on a scale of 0-14). Available nitrogen is estimated at 114 kg/ha, and available phosphorus is estimated at 5.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Baleshwar district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 56.7% (High fertility class). Soil moisture content is estimated at 18.5%. Soil pH is estimated at 5.76 (on a scale of 0-14). Available nitrogen is estimated at 109 kg/ha, and available phosphorus is estimated at 8.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bargarh district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 21.5%. Soil pH is estimated at 6.46 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Baudh district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 54.8% (Medium fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 5.42 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 8.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhadrak district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 47.2% (Medium fertility class). Soil moisture content is estimated at 27.7%. Soil pH is estimated at 5.28 (on a scale of 0-14). Available nitrogen is estimated at 133 kg/ha, and available phosphorus is estimated at 5.3 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Cuttack district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 44.0% (Medium fertility class). Soil moisture content is estimated at 17.6%. Soil pH is estimated at 6.02 (on a scale of 0-14). Available nitrogen is estimated at 119 kg/ha, and available phosphorus is estimated at 11.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Debagarh district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 41.1% (Medium fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 5.52 (on a scale of 0-14). Available nitrogen is estimated at 178 kg/ha, and available phosphorus is estimated at 9.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhenkanal district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 43.3% (Medium fertility class). Soil moisture content is estimated at 26.1%. Soil pH is estimated at 5.86 (on a scale of 0-14). Available nitrogen is estimated at 123 kg/ha, and available phosphorus is estimated at 13.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gajapati district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 42.9% (Medium fertility class). Soil moisture content is estimated at 15.5%. Soil pH is estimated at 5.74 (on a scale of 0-14). Available nitrogen is estimated at 113 kg/ha, and available phosphorus is estimated at 7.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ganjam district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 52.2% (Medium fertility class). Soil moisture content is estimated at 16.7%. Soil pH is estimated at 6.21 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red &

Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jagatsinghapur district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 51.3% (Medium fertility class). Soil moisture content is estimated at 24.3%. Soil pH is estimated at 6.49 (on a scale of 0-14). Available nitrogen is estimated at 137 kg/ha, and available phosphorus is estimated at 6.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jajapur district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 46.9% (Medium fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 5.27 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 5.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jharsuguda district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 56.3% (High fertility class). Soil moisture content is estimated at 18.4%. Soil pH is estimated at 5.06 (on a scale of 0-14). Available nitrogen is estimated at 111 kg/ha, and available phosphorus is estimated at 7.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kalahandi district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 41.3% (Medium fertility class). Soil moisture content is estimated at 15.4%. Soil pH is estimated at 5.42 (on a scale of 0-14). Available nitrogen is estimated at 121 kg/ha, and available phosphorus is estimated at 8.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kandhamal district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 43.9% (Medium fertility class). Soil moisture content is estimated at 19.4%. Soil pH is estimated at 5.49 (on a scale of 0-14). Available nitrogen is estimated at 109 kg/ha, and available phosphorus is estimated at 5.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kendrapara district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 39.6% (Medium fertility class). Soil moisture content is estimated at 24.8%. Soil pH is estimated at 6.49 (on a scale of 0-14). Available nitrogen is estimated at 202 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot

humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kendujhar district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 47.2% (Medium fertility class). Soil moisture content is estimated at 19.1%. Soil pH is estimated at 6.16 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 12.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Khordha district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 20.6%. Soil pH is estimated at 6.15 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 7.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Koraput district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 16.2%. Soil pH is estimated at 6.49 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 5.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Malkangiri district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 50.1% (Medium fertility class). Soil moisture content is estimated at 15.4%. Soil pH is estimated at 6.16 (on a scale of 0-14). Available nitrogen is estimated at 187 kg/ha, and available phosphorus is estimated at 10.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mayurbhanj district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 54.7% (Medium fertility class). Soil moisture content is estimated at 16.3%. Soil pH is estimated at 5.41 (on a scale of 0-14). Available nitrogen is estimated at 145 kg/ha, and available phosphorus is estimated at 5.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nabarangapur district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 45.3% (Medium fertility class). Soil moisture content is estimated at 18.4%. Soil pH is estimated at 5.59 (on a scale of 0-14). Available nitrogen is estimated at 146 kg/ha, and available phosphorus is estimated at

12.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nayagarh district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 41.2% (Medium fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 5.64 (on a scale of 0-14). Available nitrogen is estimated at 136 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nuapada district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 44.6% (Medium fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 5.75 (on a scale of 0-14). Available nitrogen is estimated at 116 kg/ha, and available phosphorus is estimated at 8.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Puri district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 48.0% (Medium fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 6.24 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 13.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rayagada district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 53.0% (Medium fertility class). Soil moisture content is estimated at 27.6%. Soil pH is estimated at 6.26 (on a scale of 0-14). Available nitrogen is estimated at 115 kg/ha, and available phosphorus is estimated at 9.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sambalpur district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 38.9% (Medium fertility class). Soil moisture content is estimated at 25.6%. Soil pH is estimated at 5.88 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 13.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Subarnapur district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 37.6% (Medium fertility class). Soil moisture content is estimated at 22.3%. Soil pH is estimated at 6.08 (on a scale of 0-14). Available nitrogen is estimated at 106 kg/ha, and available phosphorus is estimated at 10.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sundargarh district, Odisha. Dominant soil type: Red & Laterite Soil. Soil fertility is estimated at 53.8% (Medium fertility class). Soil moisture content is estimated at 15.9%. Soil pH is estimated at 5.77 (on a scale of 0-14). Available nitrogen is estimated at 128 kg/ha, and available phosphorus is estimated at 11.0 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1600 mm and an average temperature range of 12 to 40 degrees Celsius. Seasonal pattern: Hot humid, cyclone-prone coast, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice. [Basis: indicative regional estimate for Red & Laterite Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pondicherry

Karaikal district, Pondicherry. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 44.8% (Medium fertility class). Soil moisture content is estimated at 39.0%. Soil pH is estimated at 7.87 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 10.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1400 mm and an average temperature range of 20 to 38 degrees Celsius. Seasonal pattern: Hot humid coastal, NE monsoon Oct-Dec. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahe district, Pondicherry. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 35.5%. Soil pH is estimated at 7.33 (on a scale of 0-14). Available nitrogen is estimated at 147 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1400 mm and an average temperature range of 20 to 38 degrees Celsius. Seasonal pattern: Hot humid coastal, NE monsoon Oct-Dec. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pondicherry district, Pondicherry. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 66.0% (High fertility class). Soil moisture content is estimated at 31.3%. Soil pH is estimated at 7.85 (on a scale of 0-14). Available nitrogen is estimated at 183 kg/ha, and available phosphorus is estimated at 17.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 1200-1400 mm and an average temperature range of 20 to 38 degrees Celsius. Seasonal pattern: Hot humid coastal, NE monsoon Oct-Dec. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Yanam district, Pondicherry. Dominant soil type: Coastal Alluvial Soil. Soil fertility is estimated at 44.1% (Medium fertility class). Soil moisture content is estimated at 25.8%. Soil pH is estimated at 7.92 (on a scale of 0-14). Available nitrogen is estimated at 234 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of

1200-1400 mm and an average temperature range of 20 to 38 degrees Celsius. Seasonal pattern: Hot humid coastal, NE monsoon Oct-Dec. Crops typically suited to this soil and climate combination include: rice, coconut, areca nut and vegetables. [Basis: indicative regional estimate for Coastal Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Punjab

Amritsar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.9% (High fertility class). Soil moisture content is estimated at 24.3%. Soil pH is estimated at 7.04 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 26.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Barnala district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.3% (High fertility class). Soil moisture content is estimated at 27.9%. Soil pH is estimated at 6.59 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 22.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bathinda district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.5% (High fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 6.62 (on a scale of 0-14). Available nitrogen is estimated at 312 kg/ha, and available phosphorus is estimated at 24.0 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Faridkot district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 76.9% (High fertility class). Soil moisture content is estimated at 19.9%. Soil pH is estimated at 6.92 (on a scale of 0-14). Available nitrogen is estimated at 231 kg/ha, and available phosphorus is estimated at 20.6 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Fatehgarh Sahib district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.4% (High fertility class). Soil moisture content is estimated at 18.8%. Soil pH is estimated at 6.98 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 18.2 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional

estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Firozpur district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.9% (High fertility class). Soil moisture content is estimated at 21.7%. Soil pH is estimated at 7.32 (on a scale of 0-14). Available nitrogen is estimated at 234 kg/ha, and available phosphorus is estimated at 17.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gurdaspur district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.3% (High fertility class). Soil moisture content is estimated at 28.8%. Soil pH is estimated at 6.99 (on a scale of 0-14). Available nitrogen is estimated at 262 kg/ha, and available phosphorus is estimated at 27.5 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hoshiarpur district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.0% (High fertility class). Soil moisture content is estimated at 30.7%. Soil pH is estimated at 7.53 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 18.9 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalandhar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.5% (High fertility class). Soil moisture content is estimated at 31.8%. Soil pH is estimated at 7.1 (on a scale of 0-14). Available nitrogen is estimated at 192 kg/ha, and available phosphorus is estimated at 21.0 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kapurthala district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.2% (High fertility class). Soil moisture content is estimated at 28.5%. Soil pH is estimated at 7.16 (on a scale of 0-14). Available nitrogen is estimated at 281 kg/ha, and available phosphorus is estimated at 25.9 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ludhiana district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.7% (High fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 7.48 (on a scale of 0-14). Available nitrogen is estimated at 297 kg/ha, and available phosphorus is estimated at 25.9 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mansa district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 55.9% (High fertility class). Soil moisture content is estimated at 25.8%. Soil pH is estimated at 6.78 (on a scale of 0-14). Available nitrogen is estimated at 300 kg/ha, and available phosphorus is estimated at 22.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Moga district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.5% (High fertility class). Soil moisture content is estimated at 20.6%. Soil pH is estimated at 6.94 (on a scale of 0-14). Available nitrogen is estimated at 278 kg/ha, and available phosphorus is estimated at 15.3 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Muktsar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.3% (High fertility class). Soil moisture content is estimated at 30.4%. Soil pH is estimated at 7.49 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 16.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Patiala district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.3% (High fertility class). Soil moisture content is estimated at 27.9%. Soil pH is estimated at 7.29 (on a scale of 0-14). Available nitrogen is estimated at 223 kg/ha, and available phosphorus is estimated at 16.8 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rupnagar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.0% (High fertility class). Soil moisture content is estimated at 28.2%. Soil pH is estimated at 6.56 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 23.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination

include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sahibzada Ajit Singh Nagar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.7% (High fertility class). Soil moisture content is estimated at 28.6%. Soil pH is estimated at 7.11 (on a scale of 0-14). Available nitrogen is estimated at 248 kg/ha, and available phosphorus is estimated at 19.0 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sangrur district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.1% (High fertility class). Soil moisture content is estimated at 25.3%. Soil pH is estimated at 7.06 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 26.4 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shahid Bhagat Singh Nagar district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.9% (High fertility class). Soil moisture content is estimated at 19.1%. Soil pH is estimated at 7.0 (on a scale of 0-14). Available nitrogen is estimated at 205 kg/ha, and available phosphorus is estimated at 17.6 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tarn Taran district, Punjab. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.2% (High fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 7.49 (on a scale of 0-14). Available nitrogen is estimated at 249 kg/ha, and available phosphorus is estimated at 26.7 kg/ha. The district falls in the Semi-Arid Subtropical (Koppen BSh) climate zone, with typical annual rainfall of 400-900 mm and an average temperature range of 4 to 42 degrees Celsius. Seasonal pattern: Hot summers, cold winters, monsoon Jul-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajasthan

Ajmer district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 18.7% (Low fertility class). Soil moisture content is estimated at 3.5%. Soil pH is estimated at 8.02 (on a scale of 0-14). Available nitrogen is estimated at 45 kg/ha, and available phosphorus is estimated at 4.8 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra

(pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Alwar district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 37.7% (Medium fertility class). Soil moisture content is estimated at 4.5%. Soil pH is estimated at 7.76 (on a scale of 0-14). Available nitrogen is estimated at 80 kg/ha, and available phosphorus is estimated at 6.5 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Banswara district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 29.6% (Low fertility class). Soil moisture content is estimated at 5.6%. Soil pH is estimated at 8.57 (on a scale of 0-14). Available nitrogen is estimated at 49 kg/ha, and available phosphorus is estimated at 6.0 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Baran district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 31.8% (Low fertility class). Soil moisture content is estimated at 9.1%. Soil pH is estimated at 7.98 (on a scale of 0-14). Available nitrogen is estimated at 79 kg/ha, and available phosphorus is estimated at 3.5 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Barmer district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 21.4% (Low fertility class). Soil moisture content is estimated at 3.5%. Soil pH is estimated at 8.45 (on a scale of 0-14). Available nitrogen is estimated at 60 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bharatpur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 19.3% (Low fertility class). Soil moisture content is estimated at 9.0%. Soil pH is estimated at 8.59 (on a scale of 0-14). Available nitrogen is estimated at 46 kg/ha, and available phosphorus is estimated at 3.2 kg/ha. The

district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bhilwara district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 19.1% (Low fertility class). Soil moisture content is estimated at 9.0%. Soil pH is estimated at 7.81 (on a scale of 0-14). Available nitrogen is estimated at 78 kg/ha, and available phosphorus is estimated at 6.0 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bikaner district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 33.9% (Low fertility class). Soil moisture content is estimated at 6.4%. Soil pH is estimated at 8.15 (on a scale of 0-14). Available nitrogen is estimated at 109 kg/ha, and available phosphorus is estimated at 4.1 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bundi district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 34.6% (Low fertility class). Soil moisture content is estimated at 3.4%. Soil pH is estimated at 8.12 (on a scale of 0-14). Available nitrogen is estimated at 102 kg/ha, and available phosphorus is estimated at 7.7 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chittaurgarh district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 31.7% (Low fertility class). Soil moisture content is estimated at 4.8%. Soil pH is estimated at 8.04 (on a scale of 0-14). Available nitrogen is estimated at 65 kg/ha, and available phosphorus is estimated at 6.7 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Churu district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 27.2% (Low fertility class). Soil moisture content is estimated at 5.3%. Soil pH is estimated at 7.9 (on a scale of 0-14). Available nitrogen is estimated at 57 kg/ha, and available phosphorus is estimated at 5.6 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dausa district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 21.0% (Low fertility class). Soil moisture content is estimated at 9.1%. Soil pH is estimated at 8.28 (on a scale of 0-14). Available nitrogen is estimated at 49 kg/ha, and available phosphorus is estimated at 7.9 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dhulpur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 35.0% (Medium fertility class). Soil moisture content is estimated at 4.7%. Soil pH is estimated at 7.7 (on a scale of 0-14). Available nitrogen is estimated at 40 kg/ha, and available phosphorus is estimated at 3.3 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dungarpur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 33.2% (Low fertility class). Soil moisture content is estimated at 9.9%. Soil pH is estimated at 7.56 (on a scale of 0-14). Available nitrogen is estimated at 92 kg/ha, and available phosphorus is estimated at 3.5 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ganganagar district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 29.8% (Low fertility class). Soil moisture content is estimated at 8.3%. Soil pH is estimated at 7.97 (on a scale of 0-14). Available nitrogen is estimated at 40 kg/ha, and available phosphorus is estimated at 7.2 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis:

indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hanumangarh district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 37.8% (Medium fertility class). Soil moisture content is estimated at 4.8%. Soil pH is estimated at 7.75 (on a scale of 0-14). Available nitrogen is estimated at 48 kg/ha, and available phosphorus is estimated at 6.9 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jaipur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 29.9% (Low fertility class). Soil moisture content is estimated at 8.5%. Soil pH is estimated at 8.48 (on a scale of 0-14). Available nitrogen is estimated at 102 kg/ha, and available phosphorus is estimated at 8.6 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jaisalmer district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 22.9% (Low fertility class). Soil moisture content is estimated at 6.9%. Soil pH is estimated at 7.89 (on a scale of 0-14). Available nitrogen is estimated at 63 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalor district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 31.6% (Low fertility class). Soil moisture content is estimated at 9.8%. Soil pH is estimated at 8.04 (on a scale of 0-14). Available nitrogen is estimated at 77 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jhalawar district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 20.1% (Low fertility class). Soil moisture content is estimated at 4.7%. Soil pH is estimated at 8.19 (on a scale of 0-14). Available nitrogen is estimated at 44 kg/ha, and available phosphorus is estimated at 9.5 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold

nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jhunjhunun district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 20.5% (Low fertility class). Soil moisture content is estimated at 4.4%. Soil pH is estimated at 8.49 (on a scale of 0-14). Available nitrogen is estimated at 46 kg/ha, and available phosphorus is estimated at 9.3 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jodhpur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 32.5% (Low fertility class). Soil moisture content is estimated at 3.5%. Soil pH is estimated at 8.4 (on a scale of 0-14). Available nitrogen is estimated at 84 kg/ha, and available phosphorus is estimated at 8.3 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karauli district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 34.6% (Low fertility class). Soil moisture content is estimated at 8.0%. Soil pH is estimated at 8.26 (on a scale of 0-14). Available nitrogen is estimated at 62 kg/ha, and available phosphorus is estimated at 7.9 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kota district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 37.1% (Medium fertility class). Soil moisture content is estimated at 7.1%. Soil pH is estimated at 8.36 (on a scale of 0-14). Available nitrogen is estimated at 70 kg/ha, and available phosphorus is estimated at 8.3 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nagaur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 38.0% (Medium fertility class). Soil moisture content is estimated at 4.1%. Soil pH is estimated at 7.76 (on a scale

of 0-14). Available nitrogen is estimated at 77 kg/ha, and available phosphorus is estimated at 6.4 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pali district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 29.4% (Low fertility class). Soil moisture content is estimated at 3.3%. Soil pH is estimated at 8.03 (on a scale of 0-14). Available nitrogen is estimated at 80 kg/ha, and available phosphorus is estimated at 7.1 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pratapgarh district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 31.1% (Low fertility class). Soil moisture content is estimated at 8.0%. Soil pH is estimated at 8.33 (on a scale of 0-14). Available nitrogen is estimated at 95 kg/ha, and available phosphorus is estimated at 7.4 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rajsamand district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 28.8% (Low fertility class). Soil moisture content is estimated at 7.2%. Soil pH is estimated at 7.63 (on a scale of 0-14). Available nitrogen is estimated at 59 kg/ha, and available phosphorus is estimated at 6.7 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sawai Madhopur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 35.5% (Medium fertility class). Soil moisture content is estimated at 7.8%. Soil pH is estimated at 7.62 (on a scale of 0-14). Available nitrogen is estimated at 81 kg/ha, and available phosphorus is estimated at 4.6 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sikar district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 32.7% (Low fertility class). Soil moisture content is estimated at 8.8%. Soil pH is estimated at 8.27 (on a scale of 0-14). Available nitrogen is estimated at 81 kg/ha, and available phosphorus is estimated at 5.4 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sirohi district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 26.9% (Low fertility class). Soil moisture content is estimated at 9.0%. Soil pH is estimated at 8.44 (on a scale of 0-14). Available nitrogen is estimated at 76 kg/ha, and available phosphorus is estimated at 5.6 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tonk district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 19.0% (Low fertility class). Soil moisture content is estimated at 4.0%. Soil pH is estimated at 7.75 (on a scale of 0-14). Available nitrogen is estimated at 66 kg/ha, and available phosphorus is estimated at 9.6 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Udaipur district, Rajasthan. Dominant soil type: Arid & Desert Soil. Soil fertility is estimated at 28.1% (Low fertility class). Soil moisture content is estimated at 4.6%. Soil pH is estimated at 7.64 (on a scale of 0-14). Available nitrogen is estimated at 63 kg/ha, and available phosphorus is estimated at 8.7 kg/ha. The district falls in the Arid / Desert (Koppen BWh) climate zone, with typical annual rainfall of 100-650 mm and an average temperature range of 2 to 48 degrees Celsius. Seasonal pattern: Very hot dry summers, cold nights in winter, sparse monsoon. Crops typically suited to this soil and climate combination include: bajra (pearl millet), moth bean, guar and drought-tolerant fodder crops, mainly under irrigation; low rainfall means most cultivation depends on irrigation or drought-resistant varieties. [Basis: indicative regional estimate for Arid & Desert Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sikkim

East District district, Sikkim. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 49.6% (Medium fertility class). Soil moisture content is estimated at 34.4%. Soil pH is estimated at 6.44 (on a scale of 0-14). Available nitrogen is estimated at 133 kg/ha, and available phosphorus is estimated at 12.1 kg/ha. The district falls in the Highland / Subtropical Highland (Koppen Cwb) climate zone, with typical annual rainfall of 2000-3500 mm and an average temperature range of (-5) to 28 degrees Celsius.

Seasonal pattern: Cool to cold year-round, heavy monsoon rain. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North District district, Sikkim. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 64.0% (High fertility class). Soil moisture content is estimated at 28.4%. Soil pH is estimated at 5.09 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 14.8 kg/ha. The district falls in the Highland / Subtropical Highland (Koppen Cwb) climate zone, with typical annual rainfall of 2000-3500 mm and an average temperature range of (-5) to 28 degrees Celsius. Seasonal pattern: Cool to cold year-round, heavy monsoon rain. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South District district, Sikkim. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 53.1% (Medium fertility class). Soil moisture content is estimated at 30.8%. Soil pH is estimated at 5.38 (on a scale of 0-14). Available nitrogen is estimated at 218 kg/ha, and available phosphorus is estimated at 11.0 kg/ha. The district falls in the Highland / Subtropical Highland (Koppen Cwb) climate zone, with typical annual rainfall of 2000-3500 mm and an average temperature range of (-5) to 28 degrees Celsius. Seasonal pattern: Cool to cold year-round, heavy monsoon rain. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West District district, Sikkim. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 40.8% (Medium fertility class). Soil moisture content is estimated at 32.9%. Soil pH is estimated at 6.71 (on a scale of 0-14). Available nitrogen is estimated at 219 kg/ha, and available phosphorus is estimated at 8.8 kg/ha. The district falls in the Highland / Subtropical Highland (Koppen Cwb) climate zone, with typical annual rainfall of 2000-3500 mm and an average temperature range of (-5) to 28 degrees Celsius. Seasonal pattern: Cool to cold year-round, heavy monsoon rain. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tamil Nadu

Ariyalur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 46.2% (Medium fertility class). Soil moisture content is estimated at 25.3%. Soil pH is estimated at 7.42 (on a scale of 0-14). Available nitrogen is estimated at 198 kg/ha, and available phosphorus is estimated at 17.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chennai district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 55.6% (High fertility class). Soil moisture content is estimated at 19.0%. Soil pH is estimated at 6.36 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 12.0 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Coimbatore district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.2% (High fertility class). Soil moisture content is estimated at 15.7%. Soil pH is estimated at 6.49 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 11.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Cuddalore district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 48.8% (Medium fertility class). Soil moisture content is estimated at 20.8%. Soil pH is estimated at 6.68 (on a scale of 0-14). Available nitrogen is estimated at 208 kg/ha, and available phosphorus is estimated at 20.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dharmapuri district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 48.5% (Medium fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 7.37 (on a scale of 0-14). Available nitrogen is estimated at 231 kg/ha, and available phosphorus is estimated at 17.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dindigul district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 59.3% (High fertility class). Soil moisture content is estimated at 21.7%. Soil pH is estimated at 7.1 (on a scale of 0-14). Available nitrogen is estimated at 141 kg/ha, and available phosphorus is estimated at 21.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Erode district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 58.5% (High fertility class). Soil moisture content is estimated at 23.4%. Soil pH is estimated at 6.69 (on a scale of

0-14). Available nitrogen is estimated at 141 kg/ha, and available phosphorus is estimated at 15.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kancheepuram district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 56.7% (High fertility class). Soil moisture content is estimated at 19.2%. Soil pH is estimated at 6.64 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 15.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kanniyakumari district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 53.6% (Medium fertility class). Soil moisture content is estimated at 25.6%. Soil pH is estimated at 6.42 (on a scale of 0-14). Available nitrogen is estimated at 175 kg/ha, and available phosphorus is estimated at 12.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Karur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 57.5% (High fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 7.04 (on a scale of 0-14). Available nitrogen is estimated at 244 kg/ha, and available phosphorus is estimated at 12.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Krishnagiri district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 60.9% (High fertility class). Soil moisture content is estimated at 14.3%. Soil pH is estimated at 6.54 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Madurai district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 59.2% (High fertility class). Soil moisture content is estimated at 22.1%. Soil pH is estimated at 7.17 (on a scale of 0-14). Available nitrogen is estimated at 196 kg/ha, and available phosphorus is estimated at 16.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of

700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nagapattinam district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 54.1% (Medium fertility class). Soil moisture content is estimated at 14.5%. Soil pH is estimated at 6.33 (on a scale of 0-14). Available nitrogen is estimated at 212 kg/ha, and available phosphorus is estimated at 21.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Namakkal district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 45.4% (Medium fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 7.18 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 16.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Perambalur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 59.3% (High fertility class). Soil moisture content is estimated at 22.4%. Soil pH is estimated at 7.32 (on a scale of 0-14). Available nitrogen is estimated at 153 kg/ha, and available phosphorus is estimated at 21.0 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pudukkottai district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 45.6% (Medium fertility class). Soil moisture content is estimated at 24.0%. Soil pH is estimated at 6.3 (on a scale of 0-14). Available nitrogen is estimated at 161 kg/ha, and available phosphorus is estimated at 9.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ramanathapuram district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 55.2% (High fertility class). Soil moisture content is estimated at 18.0%. Soil pH is estimated at 6.42 (on a scale of 0-14). Available nitrogen is estimated at 189 kg/ha, and available phosphorus is estimated at 14.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate

combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Salem district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 62.7% (High fertility class). Soil moisture content is estimated at 24.7%. Soil pH is estimated at 7.05 (on a scale of 0-14). Available nitrogen is estimated at 153 kg/ha, and available phosphorus is estimated at 12.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sivaganga district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 66.0% (High fertility class). Soil moisture content is estimated at 15.1%. Soil pH is estimated at 6.73 (on a scale of 0-14). Available nitrogen is estimated at 243 kg/ha, and available phosphorus is estimated at 21.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thanjavur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 54.5% (Medium fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 222 kg/ha, and available phosphorus is estimated at 9.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

The Nilgiris district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 48.3% (Medium fertility class). Soil moisture content is estimated at 21.4%. Soil pH is estimated at 6.5 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Theni district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.4% (High fertility class). Soil moisture content is estimated at 23.8%. Soil pH is estimated at 6.84 (on a scale of 0-14). Available nitrogen is estimated at 225 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

weather reading.]

Thiruvallur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 56.4% (High fertility class). Soil moisture content is estimated at 16.4%. Soil pH is estimated at 7.32 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 20.7 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thiruvarur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.2% (High fertility class). Soil moisture content is estimated at 25.5%. Soil pH is estimated at 6.5 (on a scale of 0-14). Available nitrogen is estimated at 242 kg/ha, and available phosphorus is estimated at 16.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Thoothukkudi district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 54.7% (Medium fertility class). Soil moisture content is estimated at 20.7%. Soil pH is estimated at 6.65 (on a scale of 0-14). Available nitrogen is estimated at 171 kg/ha, and available phosphorus is estimated at 12.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tiruchirappalli district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 46.9% (Medium fertility class). Soil moisture content is estimated at 20.0%. Soil pH is estimated at 7.35 (on a scale of 0-14). Available nitrogen is estimated at 209 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tirunelveli district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 54.9% (Medium fertility class). Soil moisture content is estimated at 19.6%. Soil pH is estimated at 6.47 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 11.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tiruppur district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 61.5% (High fertility class). Soil moisture content is estimated at 23.1%. Soil pH is estimated at 6.95 (on a scale of 0-14). Available nitrogen is estimated at 234 kg/ha, and available phosphorus is estimated at 12.8 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tiruvannamalai district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 65.3% (High fertility class). Soil moisture content is estimated at 22.8%. Soil pH is estimated at 7.24 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 20.0 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Vellore district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 67.8% (High fertility class). Soil moisture content is estimated at 19.0%. Soil pH is estimated at 7.26 (on a scale of 0-14). Available nitrogen is estimated at 221 kg/ha, and available phosphorus is estimated at 14.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Viluppuram district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 66.3% (High fertility class). Soil moisture content is estimated at 18.7%. Soil pH is estimated at 6.94 (on a scale of 0-14). Available nitrogen is estimated at 151 kg/ha, and available phosphorus is estimated at 11.9 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Virudhunagar district, Tamil Nadu. Dominant soil type: Red & Black Soil. Soil fertility is estimated at 64.3% (High fertility class). Soil moisture content is estimated at 18.8%. Soil pH is estimated at 6.71 (on a scale of 0-14). Available nitrogen is estimated at 167 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 700-1200 mm and an average temperature range of 18 to 40 degrees Celsius. Seasonal pattern: Hot year-round, NE monsoon Oct-Dec is main rain season. Crops typically suited to this soil and climate combination include: groundnut, millets, cotton, pulses, oilseeds and tobacco. [Basis: indicative regional estimate for Red & Black Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tripura

Dhalai district, Tripura. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 53.5% (Medium fertility class). Soil moisture content is estimated at 30.5%. Soil pH is estimated at 4.9 (on a scale of 0-14). Available nitrogen is estimated at 157 kg/ha, and available phosphorus is estimated at 9.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 2000-2500 mm and an average temperature range of 10 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North Tripura district, Tripura. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 49.1% (Medium fertility class). Soil moisture content is estimated at 24.3%. Soil pH is estimated at 5.36 (on a scale of 0-14). Available nitrogen is estimated at 147 kg/ha, and available phosphorus is estimated at 10.2 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 2000-2500 mm and an average temperature range of 10 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South Tripura district, Tripura. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 46.5% (Medium fertility class). Soil moisture content is estimated at 24.6%. Soil pH is estimated at 5.94 (on a scale of 0-14). Available nitrogen is estimated at 174 kg/ha, and available phosphorus is estimated at 8.1 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 2000-2500 mm and an average temperature range of 10 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Tripura district, Tripura. Dominant soil type: Red & Lateritic Soil. Soil fertility is estimated at 53.4% (Medium fertility class). Soil moisture content is estimated at 30.0%. Soil pH is estimated at 5.01 (on a scale of 0-14). Available nitrogen is estimated at 158 kg/ha, and available phosphorus is estimated at 7.6 kg/ha. The district falls in the Tropical Savanna (Koppen Aw) climate zone, with typical annual rainfall of 2000-2500 mm and an average temperature range of 10 to 35 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon May-Sep. Crops typically suited to this soil and climate combination include: millets, pulses, cashew and, with irrigation, rice; the high rainfall also favours plantation crops (tea, coffee, spices) where terrain allows. [Basis: indicative regional estimate for Red & Lateritic Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttar Pradesh

Agra district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 21.3%. Soil pH is estimated at 7.37 (on a scale of 0-14). Available nitrogen is estimated at 215 kg/ha, and available phosphorus is estimated at 20.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Aligarh district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.7% (High fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 7.64 (on a scale of 0-14). Available nitrogen is estimated at 318 kg/ha, and available phosphorus is estimated at 26.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Allahabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.1% (High fertility class). Soil moisture content is estimated at 29.5%. Soil pH is estimated at 6.54 (on a scale of 0-14). Available nitrogen is estimated at 235 kg/ha, and available phosphorus is estimated at 25.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ambedkar Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.2% (High fertility class). Soil moisture content is estimated at 25.5%. Soil pH is estimated at 7.0 (on a scale of 0-14). Available nitrogen is estimated at 293 kg/ha, and available phosphorus is estimated at 23.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Auraiya district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.7% (High fertility class). Soil moisture content is estimated at 30.5%. Soil pH is estimated at 6.8 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 22.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Azamgarh district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.3% (High fertility class). Soil moisture content is estimated at 21.5%. Soil pH is estimated at 6.82 (on a scale of 0-14). Available nitrogen is estimated at 199 kg/ha, and available phosphorus is estimated at 25.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Baghpat district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.7% (High fertility class). Soil moisture content is estimated at 25.0%. Soil pH is estimated at 7.79 (on a scale of 0-14). Available nitrogen is estimated at 181 kg/ha, and available phosphorus is estimated at 20.9 kg/ha.

The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bahraich district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 67.8% (High fertility class). Soil moisture content is estimated at 29.3%. Soil pH is estimated at 6.96 (on a scale of 0-14). Available nitrogen is estimated at 237 kg/ha, and available phosphorus is estimated at 14.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ballia district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 29.0%. Soil pH is estimated at 6.68 (on a scale of 0-14). Available nitrogen is estimated at 235 kg/ha, and available phosphorus is estimated at 26.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Balrampur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 55.1% (High fertility class). Soil moisture content is estimated at 22.2%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 226 kg/ha, and available phosphorus is estimated at 16.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Banda district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.5% (High fertility class). Soil moisture content is estimated at 20.5%. Soil pH is estimated at 7.19 (on a scale of 0-14). Available nitrogen is estimated at 278 kg/ha, and available phosphorus is estimated at 21.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bara Banki district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.3% (High fertility class). Soil moisture content is estimated at 29.8%. Soil pH is estimated at 6.73 (on a scale of 0-14). Available nitrogen is estimated at 287 kg/ha, and available phosphorus is estimated at 17.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot

summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bareilly district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.6% (High fertility class). Soil moisture content is estimated at 20.9%. Soil pH is estimated at 6.74 (on a scale of 0-14). Available nitrogen is estimated at 249 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Basti district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.6% (High fertility class). Soil moisture content is estimated at 28.1%. Soil pH is estimated at 7.59 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 26.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bijnor district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.9% (High fertility class). Soil moisture content is estimated at 30.2%. Soil pH is estimated at 7.41 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 20.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Budaun district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.3% (High fertility class). Soil moisture content is estimated at 30.9%. Soil pH is estimated at 6.63 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 13.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bulandshahr district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.6% (High fertility class). Soil moisture content is estimated at 24.2%. Soil pH is estimated at 7.38 (on a scale of 0-14). Available nitrogen is estimated at 297 kg/ha, and available phosphorus is estimated at 27.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional

estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chandauli district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.0% (High fertility class). Soil moisture content is estimated at 29.2%. Soil pH is estimated at 7.65 (on a scale of 0-14). Available nitrogen is estimated at 186 kg/ha, and available phosphorus is estimated at 24.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chitrakoot district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.9% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 7.17 (on a scale of 0-14). Available nitrogen is estimated at 216 kg/ha, and available phosphorus is estimated at 14.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Deoria district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.9% (High fertility class). Soil moisture content is estimated at 19.4%. Soil pH is estimated at 7.33 (on a scale of 0-14). Available nitrogen is estimated at 229 kg/ha, and available phosphorus is estimated at 15.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Etah district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.3% (High fertility class). Soil moisture content is estimated at 27.6%. Soil pH is estimated at 7.2 (on a scale of 0-14). Available nitrogen is estimated at 310 kg/ha, and available phosphorus is estimated at 18.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Etawah district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.9% (High fertility class). Soil moisture content is estimated at 19.1%. Soil pH is estimated at 7.46 (on a scale of 0-14). Available nitrogen is estimated at 318 kg/ha, and available phosphorus is estimated at 25.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Faizabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.5% (High fertility class). Soil moisture content is estimated at 26.8%. Soil pH is estimated at 6.96 (on a scale of 0-14). Available nitrogen is estimated at 275 kg/ha, and available phosphorus is estimated at 24.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Farrukhabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 80.0% (High fertility class). Soil moisture content is estimated at 24.9%. Soil pH is estimated at 6.58 (on a scale of 0-14). Available nitrogen is estimated at 263 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Fatehpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.1% (High fertility class). Soil moisture content is estimated at 22.6%. Soil pH is estimated at 7.45 (on a scale of 0-14). Available nitrogen is estimated at 205 kg/ha, and available phosphorus is estimated at 14.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Firozabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.1% (High fertility class). Soil moisture content is estimated at 30.2%. Soil pH is estimated at 7.4 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 12.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gautam Buddha Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.0% (High fertility class). Soil moisture content is estimated at 22.0%. Soil pH is estimated at 7.28 (on a scale of 0-14). Available nitrogen is estimated at 240 kg/ha, and available phosphorus is estimated at 26.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ghaziabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.2% (High fertility class). Soil moisture content is estimated at 27.5%. Soil pH is estimated at 6.57 (on a scale of

0-14). Available nitrogen is estimated at 313 kg/ha, and available phosphorus is estimated at 27.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Ghazipur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 73.5% (High fertility class). Soil moisture content is estimated at 24.5%. Soil pH is estimated at 6.79 (on a scale of 0-14). Available nitrogen is estimated at 293 kg/ha, and available phosphorus is estimated at 23.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gonda district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.5% (High fertility class). Soil moisture content is estimated at 24.3%. Soil pH is estimated at 7.1 (on a scale of 0-14). Available nitrogen is estimated at 295 kg/ha, and available phosphorus is estimated at 22.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Gorakhpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.1% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 6.85 (on a scale of 0-14). Available nitrogen is estimated at 232 kg/ha, and available phosphorus is estimated at 12.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hamirpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.7% (High fertility class). Soil moisture content is estimated at 24.1%. Soil pH is estimated at 7.07 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 15.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hardoi district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.7% (High fertility class). Soil moisture content is estimated at 21.3%. Soil pH is estimated at 6.58 (on a scale of 0-14). Available nitrogen is estimated at 287 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot

summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalaun district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.5% (High fertility class). Soil moisture content is estimated at 24.0%. Soil pH is estimated at 7.56 (on a scale of 0-14). Available nitrogen is estimated at 224 kg/ha, and available phosphorus is estimated at 24.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jaunpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.5% (High fertility class). Soil moisture content is estimated at 27.3%. Soil pH is estimated at 7.06 (on a scale of 0-14). Available nitrogen is estimated at 284 kg/ha, and available phosphorus is estimated at 19.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jhansi district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.4% (High fertility class). Soil moisture content is estimated at 25.4%. Soil pH is estimated at 6.91 (on a scale of 0-14). Available nitrogen is estimated at 194 kg/ha, and available phosphorus is estimated at 18.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jyotiba Phule Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.1% (High fertility class). Soil moisture content is estimated at 23.7%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 310 kg/ha, and available phosphorus is estimated at 12.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kannauj district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.5% (High fertility class). Soil moisture content is estimated at 24.4%. Soil pH is estimated at 6.53 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 13.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional

estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kanpur Dehat district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.4% (High fertility class). Soil moisture content is estimated at 21.8%. Soil pH is estimated at 7.02 (on a scale of 0-14). Available nitrogen is estimated at 254 kg/ha, and available phosphorus is estimated at 12.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kanpur Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.0% (High fertility class). Soil moisture content is estimated at 19.3%. Soil pH is estimated at 7.21 (on a scale of 0-14). Available nitrogen is estimated at 226 kg/ha, and available phosphorus is estimated at 20.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kanshiram Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 60.5% (High fertility class). Soil moisture content is estimated at 18.3%. Soil pH is estimated at 7.72 (on a scale of 0-14). Available nitrogen is estimated at 195 kg/ha, and available phosphorus is estimated at 14.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kaushambi district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.7% (High fertility class). Soil moisture content is estimated at 31.7%. Soil pH is estimated at 7.3 (on a scale of 0-14). Available nitrogen is estimated at 265 kg/ha, and available phosphorus is estimated at 27.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kheri district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.2% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 6.77 (on a scale of 0-14). Available nitrogen is estimated at 267 kg/ha, and available phosphorus is estimated at 18.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kushinagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.6% (High fertility class). Soil moisture content is estimated at 31.0%. Soil pH is estimated at 6.69 (on a scale of 0-14). Available nitrogen is estimated at 319 kg/ha, and available phosphorus is estimated at 18.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lalitpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.2% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 309 kg/ha, and available phosphorus is estimated at 21.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Lucknow district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.6% (High fertility class). Soil moisture content is estimated at 20.1%. Soil pH is estimated at 7.19 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 19.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahamaya Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.9% (High fertility class). Soil moisture content is estimated at 21.4%. Soil pH is estimated at 7.61 (on a scale of 0-14). Available nitrogen is estimated at 184 kg/ha, and available phosphorus is estimated at 23.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahoba district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 80.0% (High fertility class). Soil moisture content is estimated at 18.1%. Soil pH is estimated at 6.71 (on a scale of 0-14). Available nitrogen is estimated at 274 kg/ha, and available phosphorus is estimated at 13.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mahrajganj district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.7% (High fertility class). Soil moisture content is estimated at 27.6%. Soil pH is estimated at 7.26 (on a scale of

0-14). Available nitrogen is estimated at 231 kg/ha, and available phosphorus is estimated at 23.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mainpuri district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 69.9% (High fertility class). Soil moisture content is estimated at 20.9%. Soil pH is estimated at 7.23 (on a scale of 0-14). Available nitrogen is estimated at 303 kg/ha, and available phosphorus is estimated at 22.7 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mathura district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.8% (High fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 6.61 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 12.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mau district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.7% (High fertility class). Soil moisture content is estimated at 21.1%. Soil pH is estimated at 7.56 (on a scale of 0-14). Available nitrogen is estimated at 203 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Meerut district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.5% (High fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 7.54 (on a scale of 0-14). Available nitrogen is estimated at 314 kg/ha, and available phosphorus is estimated at 26.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Mirzapur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.8% (High fertility class). Soil moisture content is estimated at 28.9%. Soil pH is estimated at 6.69 (on a scale of 0-14). Available nitrogen is estimated at 307 kg/ha, and available phosphorus is estimated at 20.3 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of

700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Moradabad district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.4% (High fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 6.52 (on a scale of 0-14). Available nitrogen is estimated at 214 kg/ha, and available phosphorus is estimated at 26.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Muzaffarnagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.9% (High fertility class). Soil moisture content is estimated at 31.6%. Soil pH is estimated at 6.63 (on a scale of 0-14). Available nitrogen is estimated at 262 kg/ha, and available phosphorus is estimated at 16.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pilibhit district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 68.6% (High fertility class). Soil moisture content is estimated at 22.5%. Soil pH is estimated at 6.87 (on a scale of 0-14). Available nitrogen is estimated at 260 kg/ha, and available phosphorus is estimated at 14.9 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pratapgarh district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.7% (High fertility class). Soil moisture content is estimated at 27.1%. Soil pH is estimated at 6.67 (on a scale of 0-14). Available nitrogen is estimated at 310 kg/ha, and available phosphorus is estimated at 20.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rae Bareilly district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.5% (High fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 7.42 (on a scale of 0-14). Available nitrogen is estimated at 280 kg/ha, and available phosphorus is estimated at 26.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination

include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rampur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.3% (High fertility class). Soil moisture content is estimated at 21.9%. Soil pH is estimated at 6.63 (on a scale of 0-14). Available nitrogen is estimated at 264 kg/ha, and available phosphorus is estimated at 13.1 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Saharanpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 61.0% (High fertility class). Soil moisture content is estimated at 18.8%. Soil pH is estimated at 7.21 (on a scale of 0-14). Available nitrogen is estimated at 202 kg/ha, and available phosphorus is estimated at 21.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sant Kabir Nagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.8% (High fertility class). Soil moisture content is estimated at 29.5%. Soil pH is estimated at 6.9 (on a scale of 0-14). Available nitrogen is estimated at 256 kg/ha, and available phosphorus is estimated at 21.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sant Ravidas Nagar (Bhadohi) district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 66.8% (High fertility class). Soil moisture content is estimated at 27.2%. Soil pH is estimated at 6.77 (on a scale of 0-14). Available nitrogen is estimated at 317 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Shahjahanpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 75.7% (High fertility class). Soil moisture content is estimated at 26.3%. Soil pH is estimated at 6.82 (on a scale of 0-14). Available nitrogen is estimated at 284 kg/ha, and available phosphorus is estimated at 13.4 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

reading.]

Shrawasti district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.4% (High fertility class). Soil moisture content is estimated at 26.3%. Soil pH is estimated at 6.64 (on a scale of 0-14). Available nitrogen is estimated at 191 kg/ha, and available phosphorus is estimated at 24.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Siddharthnagar district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.3% (High fertility class). Soil moisture content is estimated at 30.5%. Soil pH is estimated at 7.49 (on a scale of 0-14). Available nitrogen is estimated at 277 kg/ha, and available phosphorus is estimated at 16.5 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sitapur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.7% (High fertility class). Soil moisture content is estimated at 31.3%. Soil pH is estimated at 6.62 (on a scale of 0-14). Available nitrogen is estimated at 225 kg/ha, and available phosphorus is estimated at 15.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sonbhadra district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.8% (High fertility class). Soil moisture content is estimated at 28.4%. Soil pH is estimated at 7.12 (on a scale of 0-14). Available nitrogen is estimated at 259 kg/ha, and available phosphorus is estimated at 17.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Sultanpur district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 58.5% (High fertility class). Soil moisture content is estimated at 21.7%. Soil pH is estimated at 7.62 (on a scale of 0-14). Available nitrogen is estimated at 264 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Unnao district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.7% (High fertility class). Soil moisture content is estimated at 18.9%. Soil pH is estimated at 6.83 (on a scale of 0-14). Available nitrogen is estimated at 212 kg/ha, and available phosphorus is estimated at 27.8 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Varanasi district, Uttar Pradesh. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.7% (High fertility class). Soil moisture content is estimated at 26.6%. Soil pH is estimated at 6.51 (on a scale of 0-14). Available nitrogen is estimated at 205 kg/ha, and available phosphorus is estimated at 22.2 kg/ha. The district falls in the Humid Subtropical (Koppen Cwa) climate zone, with typical annual rainfall of 700-1400 mm and an average temperature range of 4 to 45 degrees Celsius. Seasonal pattern: Hot summers, cool winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttarakhand

Almora district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 49.5% (Medium fertility class). Soil moisture content is estimated at 36.8%. Soil pH is estimated at 6.38 (on a scale of 0-14). Available nitrogen is estimated at 173 kg/ha, and available phosphorus is estimated at 13.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bageshwar district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 64.4% (High fertility class). Soil moisture content is estimated at 29.4%. Soil pH is estimated at 6.16 (on a scale of 0-14). Available nitrogen is estimated at 230 kg/ha, and available phosphorus is estimated at 8.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Chamoli district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 62.2% (High fertility class). Soil moisture content is estimated at 23.6%. Soil pH is estimated at 6.17 (on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 12.1 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Champawat district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 52.4% (Medium fertility class). Soil moisture content is estimated at 31.9%. Soil pH is estimated at 5.27 (on a scale of 0-14). Available nitrogen is estimated at 145 kg/ha, and available phosphorus is estimated at 8.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dehradun district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 54.9% (Medium fertility class). Soil moisture content is estimated at 29.1%. Soil pH is estimated at 5.44 (on a scale of 0-14). Available nitrogen is estimated at 225 kg/ha, and available phosphorus is estimated at 9.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Garhwal district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 51.2% (Medium fertility class). Soil moisture content is estimated at 28.6%. Soil pH is estimated at 5.33 (on a scale of 0-14). Available nitrogen is estimated at 207 kg/ha, and available phosphorus is estimated at 10.5 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hardwar district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 40.4% (Medium fertility class). Soil moisture content is estimated at 37.4%. Soil pH is estimated at 6.47 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 14.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nainital district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 62.7% (High fertility class). Soil moisture content is estimated at 27.4%. Soil pH is estimated at 6.1 (on a scale of 0-14). Available nitrogen is estimated at 193 kg/ha, and available phosphorus is estimated at 8.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Pithoragarh district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 45.5% (Medium fertility class). Soil moisture content is estimated at 23.8%. Soil pH is estimated at 6.19

(on a scale of 0-14). Available nitrogen is estimated at 213 kg/ha, and available phosphorus is estimated at 13.6 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Rudraprayag district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 47.9% (Medium fertility class). Soil moisture content is estimated at 25.3%. Soil pH is estimated at 5.82 (on a scale of 0-14). Available nitrogen is estimated at 185 kg/ha, and available phosphorus is estimated at 15.2 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Tehri Garhwal district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 61.9% (High fertility class). Soil moisture content is estimated at 36.7%. Soil pH is estimated at 6.53 (on a scale of 0-14). Available nitrogen is estimated at 235 kg/ha, and available phosphorus is estimated at 10.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Udham Singh Nagar district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 41.2% (Medium fertility class). Soil moisture content is estimated at 23.3%. Soil pH is estimated at 5.7 (on a scale of 0-14). Available nitrogen is estimated at 150 kg/ha, and available phosphorus is estimated at 7.7 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttarkashi district, Uttarakhand. Dominant soil type: Mountain & Forest Soil. Soil fertility is estimated at 63.8% (High fertility class). Soil moisture content is estimated at 25.1%. Soil pH is estimated at 6.73 (on a scale of 0-14). Available nitrogen is estimated at 211 kg/ha, and available phosphorus is estimated at 12.4 kg/ha. The district falls in the Humid Subtropical / Highland (Koppen Cwa/ET) climate zone, with typical annual rainfall of 1000-2000 mm and an average temperature range of 0 to 35 degrees Celsius. Seasonal pattern: Cold winters in hills, warm plains, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: temperate fruits (apple, apricot), potato, maize and terraced rice/wheat. [Basis: indicative regional estimate for Mountain & Forest Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

West Bengal

Bankura district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.9% (High fertility class). Soil moisture content is estimated at 22.6%. Soil pH is estimated at 7.34 (on a scale of 0-14). Available nitrogen is estimated at 206 kg/ha, and available phosphorus is estimated at 27.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Bardhaman district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.5% (High fertility class). Soil moisture content is estimated at 21.6%. Soil pH is estimated at 7.42 (on a scale of 0-14). Available nitrogen is estimated at 271 kg/ha, and available phosphorus is estimated at 19.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Birbhum district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 70.8% (High fertility class). Soil moisture content is estimated at 20.2%. Soil pH is estimated at 7.38 (on a scale of 0-14). Available nitrogen is estimated at 217 kg/ha, and available phosphorus is estimated at 16.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Dakshin Dinajpur district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.1% (High fertility class). Soil moisture content is estimated at 22.7%. Soil pH is estimated at 6.93 (on a scale of 0-14). Available nitrogen is estimated at 243 kg/ha, and available phosphorus is estimated at 15.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Darjiling district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.8% (High fertility class). Soil moisture content is estimated at 25.6%. Soil pH is estimated at 6.67 (on a scale of 0-14). Available nitrogen is estimated at 276 kg/ha, and available phosphorus is estimated at 27.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Haora district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 65.7% (High fertility class). Soil moisture content is estimated at 30.6%. Soil pH is estimated at 6.63 (on a scale of

0-14). Available nitrogen is estimated at 186 kg/ha, and available phosphorus is estimated at 23.5 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Hugli district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 74.0% (High fertility class). Soil moisture content is estimated at 29.8%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 288 kg/ha, and available phosphorus is estimated at 18.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Jalpaiguri district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 62.6% (High fertility class). Soil moisture content is estimated at 20.4%. Soil pH is estimated at 7.79 (on a scale of 0-14). Available nitrogen is estimated at 188 kg/ha, and available phosphorus is estimated at 25.2 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Koch Bihar district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 71.0% (High fertility class). Soil moisture content is estimated at 28.3%. Soil pH is estimated at 7.43 (on a scale of 0-14). Available nitrogen is estimated at 300 kg/ha, and available phosphorus is estimated at 14.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Kolkata district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 55.0% (High fertility class). Soil moisture content is estimated at 26.4%. Soil pH is estimated at 6.96 (on a scale of 0-14). Available nitrogen is estimated at 197 kg/ha, and available phosphorus is estimated at 15.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Maldah district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 63.3% (High fertility class). Soil moisture content is estimated at 19.0%. Soil pH is estimated at 6.57 (on a scale of 0-14). Available nitrogen is estimated at 308 kg/ha, and available phosphorus is estimated at 17.0 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of

1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Murshidabad district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.6% (High fertility class). Soil moisture content is estimated at 31.5%. Soil pH is estimated at 6.69 (on a scale of 0-14). Available nitrogen is estimated at 280 kg/ha, and available phosphorus is estimated at 21.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Nadia district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 72.3% (High fertility class). Soil moisture content is estimated at 19.3%. Soil pH is estimated at 6.75 (on a scale of 0-14). Available nitrogen is estimated at 190 kg/ha, and available phosphorus is estimated at 21.7 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

North Twenty Four Parganas district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 79.7% (High fertility class). Soil moisture content is estimated at 27.4%. Soil pH is estimated at 7.42 (on a scale of 0-14). Available nitrogen is estimated at 317 kg/ha, and available phosphorus is estimated at 22.8 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Paschim Medinipur district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 56.0% (High fertility class). Soil moisture content is estimated at 30.1%. Soil pH is estimated at 6.97 (on a scale of 0-14). Available nitrogen is estimated at 180 kg/ha, and available phosphorus is estimated at 20.1 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Purba Medinipur district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 78.4% (High fertility class). Soil moisture content is estimated at 21.6%. Soil pH is estimated at 7.14 (on a scale of 0-14). Available nitrogen is estimated at 204 kg/ha, and available phosphorus is estimated at 12.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate

combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Puruliya district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 57.0% (High fertility class). Soil moisture content is estimated at 28.2%. Soil pH is estimated at 7.62 (on a scale of 0-14). Available nitrogen is estimated at 200 kg/ha, and available phosphorus is estimated at 16.6 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

South Twenty Four Parganas district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 59.9% (High fertility class). Soil moisture content is estimated at 18.4%. Soil pH is estimated at 7.22 (on a scale of 0-14). Available nitrogen is estimated at 277 kg/ha, and available phosphorus is estimated at 22.4 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]

Uttar Dinajpur district, West Bengal. Dominant soil type: Alluvial Soil. Soil fertility is estimated at 64.7% (High fertility class). Soil moisture content is estimated at 25.8%. Soil pH is estimated at 6.62 (on a scale of 0-14). Available nitrogen is estimated at 201 kg/ha, and available phosphorus is estimated at 26.9 kg/ha. The district falls in the Tropical Wet & Dry (Koppen Aw/Cwa) climate zone, with typical annual rainfall of 1200-3000 mm and an average temperature range of 9 to 40 degrees Celsius. Seasonal pattern: Hot humid summers, mild winters, monsoon Jun-Sep. Crops typically suited to this soil and climate combination include: wheat, rice, sugarcane, maize, pulses and a wide range of vegetables. [Basis: indicative regional estimate for Alluvial Soil soil and state-level climatology; not an individual lab measurement or live weather reading.]